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JAIN (DEEMED-TO-BE) UNIVERSITY

B. Tech CSE-AIML

PROJECT REPORT ON

"Diabetes Prediction Using Machine Learning"

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BACHELOR OF TECHNOLOGY IN

COMPUTER SCIENCE AND AIML

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This is to certify that the project entitled " **Diabetes Prediction Using Machine Learning** " is a bonafide record of the work done by Prisca. S (23btrcl068), Ruhee Zainab (23btrcl159) of the B.Tech (Artificial Intelligence and Machine Learning) programme, Jain University, during the academic year 2023–2026.

The project has been carried out under my guidance and supervision. The work presented is original and has not been submitted, either in part or in full, for the award of any other degree or diploma at this or any other institution.

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ABSTRACT

Diabetes mellitus is a chronic metabolic disorder that poses a significant public health challenge globally, affecting millions of individuals across all age groups. Early identification of individuals at risk of developing diabetes is critical for enabling timely preventive interventions and reducing the long-term burden on healthcare systems. This project presents the design and implementation of a Lifestyle-Based Diabetes Risk Prediction and Early Warning System using Machine Learning, aimed at providing individuals with an accessible, accurate, and personalised tool for assessing their diabetes risk.

The system utilises the Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset published by the Centers for Disease Control and Prevention (CDC), which contains health and lifestyle data collected from approximately 253,000 Americans. A Random Forest Classifier is trained on 21 demographic, clinical, and lifestyle features to classify individuals into three categories: No Diabetes, Pre-Diabetes, and Diabetes. The trained model achieves an overall accuracy of approximately 84.79 percent on held-out test data.

A key contribution of this system is the conversion of raw classification probabilities into a continuous, interpretable Risk Score ranging from 0 to 100, computed as a weighted sum of the pre-diabetes and diabetes probabilities. Individuals are subsequently categorised into Low Risk (score below 30), Medium Risk (score between 30 and 60), and High Risk (score above 60). A Lifestyle Simulation Module further enables users to understand how progressive improvements in their lifestyle, such as quitting smoking, increasing physical activity, and reducing body mass index, correspond to measurable reductions in their risk score, thereby motivating preventive action.

The system is implemented as a full-stack web application. A Flask-based RESTful API serves as the backend, exposing endpoints for prediction, simulation, and feature importance retrieval. A professionally designed single-page frontend, developed using HTML, CSS, and JavaScript, provides an intuitive interface for users to enter their health data and receive instant results. An optional Wearable Integration module accepts smartwatch sensor data, including step count, resting heart rate, sleep hours, and calories burned, and maps these readings to model-compatible features, forming a hybrid prediction pipeline.

The project demonstrates that machine learning, when applied to publicly available health survey data, can serve as an effective and scalable tool for population-level diabetes early warning and prevention.

The enhanced system also includes a derived cardiac analysis component that infers cardiovascular stress patterns from wearable data, and an intelligent recommendation engine that provides structured, personalised health advice based on both diabetes risk and cardiac condition.

Keywords: Diabetes Prediction, Random Forest, BRFSS, Risk Score, Lifestyle Simulation, Wearable Integration, Flask, Early Warning System.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AIML	Artificial Intelligence and Machine Learning
API	Application Programming Interface
BMI	Body Mass Index
BRFSS	Behavioral Risk Factor Surveillance System
CDC	Centers for Disease Control and Prevention
CSS	Cascading Style Sheets
DT	Decision Tree
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
JS	JavaScript
JSON	JavaScript Object Notation
ML	Machine Learning
RF	Random Forest
REST	Representational State Transfer
ROC	Receiver Operating Characteristic
SVM	Support Vector Machine
UI	User Interface
WHO	World Health Organization

CHAPTER 1 : INTRODUCTION

1.1 Background and Motivation

Diabetes mellitus is one of the fastest-growing chronic diseases of the modern era. According to the World Health Organization (WHO), the global prevalence of diabetes among adults has risen dramatically over the past four decades, and projections suggest that the number of individuals living with diabetes will continue to increase substantially in the coming years. In India alone, the International Diabetes Federation estimates that over 100 million people currently have diabetes, with many more remaining undiagnosed.

The disease is broadly classified into Type 1 Diabetes, which is characterised by an absolute deficiency of insulin due to autoimmune destruction of pancreatic beta cells, and Type 2 Diabetes, which arises from a combination of insulin resistance and progressive beta-cell dysfunction. Type 2 Diabetes accounts for approximately 90 to 95 percent of all diabetes cases worldwide and is strongly associated with modifiable lifestyle factors such as physical inactivity, unhealthy diet, obesity, smoking, and heavy alcohol consumption. A third intermediate condition known as Pre-Diabetes, characterised by elevated blood glucose levels that do not yet meet the diagnostic threshold for diabetes, represents a critical preventive window during which lifestyle interventions can halt or reverse disease progression.

Despite the availability of clinical screening tools, a significant proportion of high-risk individuals remain unaware of their condition until diabetes has already developed and secondary complications, including cardiovascular disease, nephropathy, retinopathy, and neuropathy, have begun to manifest. This diagnostic delay is attributable to the asymptomatic nature of early-stage diabetes and pre-diabetes, combined with limited access to regular medical screening in resource-constrained settings.

The growing availability of large-scale health survey datasets, combined with advances in machine learning, presents a compelling opportunity to develop automated, accessible, and cost-effective risk prediction systems. Such systems can supplement traditional clinical screening, enable community-level risk stratification, and empower individuals to take proactive steps toward disease prevention.

1.2 Problem Statement

Existing diabetes screening mechanisms are primarily reactive in nature, relying on blood glucose tests administered in clinical settings after symptoms or risk factors have become apparent. These approaches are not scalable for large-scale population screening, particularly in low-resource environments. Furthermore, even when risk is identified, individuals are often presented with a binary diagnosis rather than a nuanced, personalised assessment that communicates the degree of risk and the lifestyle changes that could reduce it.

There is a clear need for a system that can: (i) predict diabetes risk using non-invasive, self-reported health and lifestyle data; (ii) provide a continuous, interpretable risk score rather than a simple binary classification; (iii) simulate the projected improvement in risk resulting from specific lifestyle changes; and (iv) integrate objective wearable sensor data to supplement subjective self-reports, thereby improving the accuracy and credibility of the prediction.

1.3 Scope of the Project

This project focuses on the development of a machine learning-based diabetes risk prediction system that operates on lifestyle, demographic, and clinical features derived from the BRFSS 2015 dataset. The scope encompasses the following components:

- Design and training of a Random Forest classification model for three-class diabetes risk prediction.
- Development of a continuous risk scoring mechanism that converts raw classification probabilities into a score ranging from 0 to 100.
- Implementation of a Lifestyle Simulation Module that demonstrates the progressive impact of health improvements on the risk score across four defined stages.
- Construction of a RESTful API backend using Flask to expose model predictions, simulations, and feature importance data.
- Development of a single-page web application frontend providing an intuitive health assessment form and dynamic result visualisation.
- Optional integration of wearable sensor data from smartwatch devices, including step counts, resting heart rate, sleep hours, and calories burned, mapped to model-compatible features.

The project does not encompass clinical validation, integration with electronic health records, real-time physiological monitoring, or deep learning-based approaches.

1.4 Objectives of the Project

The primary objectives of this project are as follows:

1. To analyse the BRFSS 2015 health survey dataset and identify the most significant lifestyle and clinical features contributing to diabetes risk.
2. To train and evaluate a Random Forest classifier capable of predicting the probability of an individual belonging to the No Diabetes, Pre-Diabetes, or Diabetes class.
3. To develop a Risk Scoring Formula that translates model output probabilities into a clinically interpretable risk index on a scale of 0 to 100, with associated risk categories.
4. To implement a Lifestyle Simulation Module that quantifies the expected reduction in risk score resulting from four progressive stages of lifestyle improvement.
5. To build a full-stack web application integrating a Flask API backend and a responsive HTML/CSS/JavaScript frontend for real-world usability.
6. To develop a Wearable Data Integration module that accepts smartwatch sensor data and maps it to model features, enabling a hybrid prediction pipeline.
7. To evaluate the model using standard classification performance metrics, including accuracy, precision, recall, F1-score, and confusion matrix analysis.

CHAPTER 2: LITERATURE SURVEY

2.1 Machine Learning in Healthcare

The application of machine learning to healthcare problems has grown substantially over the past decade, driven by the increasing availability of large clinical and epidemiological datasets, improvements in computational resources, and advances in algorithmic methods. Supervised learning techniques, including decision trees, support vector machines, logistic regression, and ensemble methods such as random forests and gradient boosting, have been widely employed for disease diagnosis, patient risk stratification, and treatment outcome prediction.

Obermeyer and Emanuel (2016) highlighted the transformative potential of machine learning in clinical decision support, noting that algorithms trained on electronic health record data could outperform traditional statistical models in predicting patient deterioration. Rajpurkar et al. (2017) demonstrated the capability of deep learning models in detecting pneumonia from chest radiographs with cardiologist-level accuracy, marking a milestone in AI-assisted diagnostics. Within the domain of chronic disease management, numerous studies have applied ensemble learning methods to predict the onset and progression of conditions such as cardiovascular disease, chronic kidney disease, and diabetes.

Random forests, introduced by Breiman (2001), have proven particularly effective in healthcare classification tasks owing to their robustness to overfitting, ability to handle high-dimensional mixed-type feature spaces, and built-in feature importance estimation. These properties make them well-suited for health survey datasets such as BRFSS, which contain a combination of binary, ordinal, and continuous variables.

2.2 Diabetes Prediction Systems

The prediction of diabetes using machine learning has been an active research area for over a decade. Early studies primarily utilised the Pima Indians Diabetes Dataset, a smaller dataset comprising 768 instances with eight clinical features, to benchmark classification algorithms. Kavakiotis et al. (2017) conducted a comprehensive review of machine learning and data mining techniques applied to diabetes research, documenting the dominance of support vector machines and neural networks in the literature, while also

noting that random forest and decision tree-based approaches consistently delivered competitive performance.

Sisodia and Sisodia (2018) applied naive Bayes, decision tree, and support vector machine classifiers to the Pima dataset and reported classification accuracies ranging from 72 to 76 percent. Zou et al. (2018) compared multiple machine learning approaches on a clinical dataset from Luzhou, China, and found that random forests achieved the highest AUC-ROC score among the evaluated models. More recent studies have leveraged the larger BRFSS dataset, which provides a richer and more diverse representation of real-world health behaviours, to develop population-scale prediction models.

Xue et al. (2021) utilised BRFSS data to develop a gradient-boosted ensemble model for diabetes risk prediction, reporting an accuracy of approximately 86 percent and emphasising the importance of lifestyle features, particularly BMI, general health rating, and physical activity, as dominant predictors. Islam et al. (2020) examined the utility of hybrid models that combine clinical features with self-reported lifestyle indicators, demonstrating that the inclusion of behavioural variables significantly improved model discrimination.

2.3 Wearable Data in Healthcare

The widespread adoption of consumer-grade wearable devices, including fitness trackers and smartwatches, has opened new avenues for continuous, objective health monitoring. Devices such as Fitbit, Apple Watch, Garmin, and Samsung Galaxy Watch routinely collect physiological metrics including step count, heart rate, sleep duration, blood oxygen saturation, and estimated caloric expenditure.

Dunn et al. (2018) demonstrated in a landmark study that wearable-derived data could detect pre-diabetes and insulin resistance by identifying deviations in continuous glucose monitoring patterns, even in the absence of traditional clinical tests. Bent et al. (2020) showed that wearable physiological signals, when combined with demographic information, could predict metabolic health indicators with statistically significant accuracy.

In the context of diabetes prediction, wearable data offers a complementary, objective dimension to self-reported survey responses. Physical inactivity, measured as daily step count below the recommended 7,500 steps per day, elevated resting heart rate

indicative of cardiovascular stress, and poor sleep quality as reflected by insufficient sleep hours are all established risk factors for Type 2 Diabetes. The integration of wearable metrics into a machine learning prediction pipeline represents a clinically meaningful enhancement over systems relying exclusively on self-reported data.

2.4 Research Gap

A review of the existing literature reveals several notable gaps that the present project seeks to address. First, the majority of published diabetes prediction systems produce a binary output, classifying individuals as diabetic or non-diabetic, without providing a nuanced, continuous risk score that conveys gradations of risk. Second, most systems do not incorporate a simulation component that enables users to understand the quantitative impact of specific lifestyle modifications on their predicted risk, limiting their utility as motivational tools for behaviour change.

Third, the integration of wearable sensor data with survey-based prediction models remains relatively unexplored in publicly available implementations, despite the growing availability of such data among health-conscious consumers. Fourth, few existing systems are deployed as accessible, user-friendly web applications; the prevailing implementations remain as standalone research prototypes without practical interfaces for non-technical users. The present project directly addresses all four of these gaps.

CHAPTER 3: EXISTING SYSTEM

3.1 Current Methods

Current approaches to diabetes screening and risk assessment can be broadly categorised into three groups: clinical laboratory-based methods, clinical questionnaire-based tools, and computational prediction models.

3.1.1 Clinical Laboratory-Based Screening

The gold standard for diabetes diagnosis involves measuring fasting plasma glucose, two-hour plasma glucose during an oral glucose tolerance test, or glycated haemoglobin (HbA1c) concentration. While these tests are highly accurate, they require venepuncture, laboratory infrastructure, and trained clinical personnel, rendering them unsuitable for routine mass screening in primary care or community settings.

3.1.2 Questionnaire-Based Risk Tools

Several validated clinical risk tools have been developed to estimate diabetes risk from self-reported information without requiring laboratory tests. The Finnish Diabetes Risk Score (FINDRISC), developed by Lindstrom and Tuomilehto (2003), uses eight items including age, BMI, waist circumference, physical activity, dietary habits, and family history to classify individuals into five risk categories. The American Diabetes Association (ADA) Risk Test is a simpler seven-item questionnaire that provides a binary risk classification. While these tools are practical and widely used in clinical practice, they produce coarse, categorical outputs and do not leverage the predictive power of modern machine learning algorithms.

3.1.3 Existing Computational Models

A range of machine learning-based diabetes prediction systems have been described in the academic literature and made available as online tools. Notable examples include DiabetesMine, which uses a logistic regression model trained on the Pima Indians dataset, and several web-based calculators that apply decision tree classifiers. These tools typically present a binary or ternary risk classification based on a limited set of clinical variables, without providing a continuous risk score, a lifestyle improvement simulation, or integration with wearable data.

3.2 Limitations of Existing Systems

A systematic analysis of existing diabetes prediction and risk assessment systems reveals the following significant limitations:

- Binary Classification Output: Most systems provide a simple binary diabetic or non-diabetic classification, which fails to capture the spectrum of risk that is clinically and behaviourally meaningful. The absence of a continuous score prevents users from understanding how close they are to a risk threshold or how much improvement they have achieved.
- Absence of Lifestyle Simulation: No widely available prediction tool incorporates a module that simulates the projected change in risk score resulting from defined lifestyle improvements. This omission represents a significant missed opportunity for motivating preventive behaviour change.
- Limited Feature Sets: Many existing tools rely on a small number of clinical variables, typically fewer than ten, and do not capture the breadth of lifestyle and socioeconomic factors that are known to influence diabetes risk, such as mental health status, income level, education, and access to healthcare.
- No Wearable Integration: Existing systems do not accept or process data from wearable devices, despite the widespread adoption of fitness trackers and the objective, continuous nature of the health data they provide.
- Poor Accessibility and User Experience: The majority of existing computational models are implemented as research prototypes without user-friendly web interfaces, making them inaccessible to the general public.
- Lack of Explainability: Existing tools rarely provide users with information about which features contributed most to their risk assessment, limiting the actionability of the results.

CHAPTER 4: PROPOSED SYSTEM

4.1 System Overview

The proposed system is a full-stack, machine learning-powered diabetes risk prediction and early warning platform that addresses each of the limitations identified in the existing systems. At its core, the system trains a Random Forest Classifier on the BRFSS 2015 dataset to classify individuals into three classes: No Diabetes, Pre-Diabetes, and Diabetes. The raw classification probabilities produced by the model are transformed into a continuous Risk Score on a scale from 0 to 100, which is subsequently mapped to one of three risk categories: Low Risk, Medium Risk, or High Risk. A Lifestyle Simulation Module enables users to project the impact of progressive lifestyle improvements on their risk score, providing concrete and personalised motivation for behaviour change. An optional Wearable Integration module accepts smartwatch sensor data and maps it to model-compatible features, creating a hybrid prediction pipeline.

The system is deployed as a web application comprising a Python Flask backend API and a responsive HTML/CSS/JavaScript frontend, making it accessible to any user with a web browser and an internet connection.

4.2 Architecture Description

The system architecture follows a classic three-tier client-server model, comprising a Presentation Layer, an Application Logic Layer, and a Data and Model Layer.

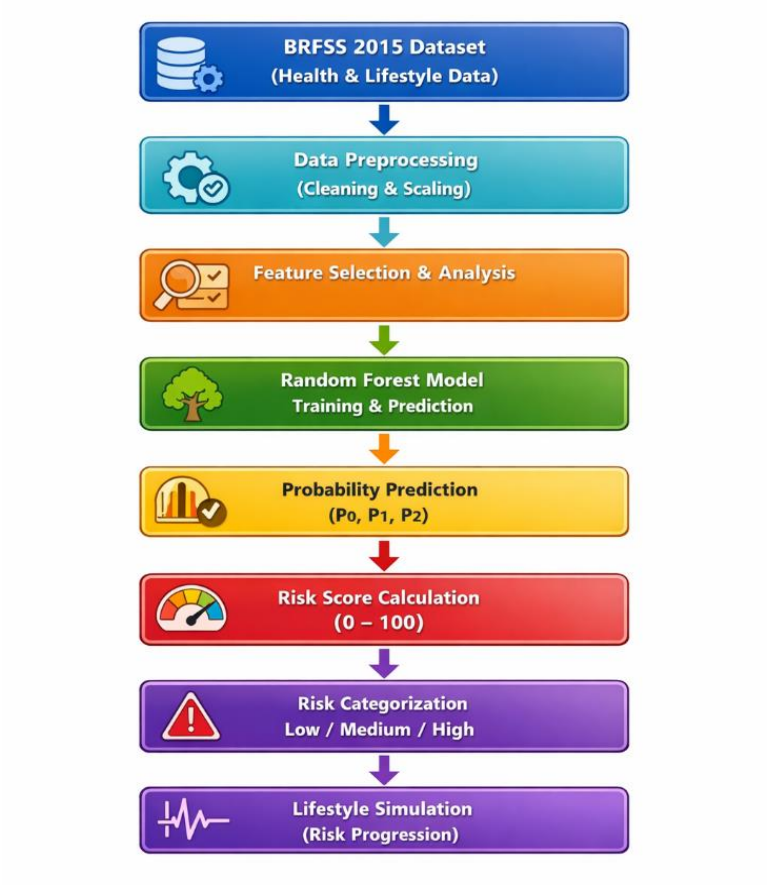


FIG 1 : ARTITECTURE DIAGRAM

Presentation Layer

The Presentation Layer consists of a single-page web application developed in HTML5, CSS3, and vanilla JavaScript. It provides a health assessment form collecting all 21 model input features, an optional wearable data input panel, animated result visualisations including a risk score ring and probability bars, a lifestyle simulation bar chart, and a feature importance display. The layer communicates exclusively with the Application Logic Layer via asynchronous HTTP requests using the JavaScript Fetch API.

Application Logic Layer

The Application Logic Layer is implemented as a Flask RESTful API hosted on a local Python server. It exposes five endpoints: a training endpoint for initial model training and persistence, a prediction endpoint for risk score computation, a simulation endpoint for lifestyle improvement analysis, a feature importance endpoint for model interpretability, and a wearable endpoint for hybrid input processing. The layer handles all data validation, feature ordering, scaling, and model inference operations. The enhanced system also

includes a derived cardiac analysis component that infers cardiovascular stress patterns from wearable data, and an intelligent recommendation engine that provides structured, personalised health advice based on both diabetes risk and cardiac condition.

Data and Model Layer

The Data and Model Layer comprises the BRFSS 2015 training dataset, a serialised Random Forest model stored as a joblib pickle file, and a serialised StandardScaler object used to normalise input features prior to inference. Once trained, the model is loaded into memory at server startup and reused for all subsequent prediction requests, ensuring efficient response times.

4.3 Data Flow

The end-to-end data flow through the system proceeds as follows. A user accesses the web application and completes the health assessment form, optionally supplementing their responses with wearable sensor readings. Upon form submission, the JavaScript frontend assembles a JSON payload containing the user's feature values and transmits it to the appropriate Flask endpoint via an HTTP POST request.

At the backend, the received JSON is parsed and the feature values are extracted in the correct column order required by the trained model. If wearable data is present, the wearable mapping function translates sensor readings into model-compatible feature values, which are merged with the manually entered data, with manual entries taking precedence where overlap exists. The merged feature vector is passed through the StandardScaler for normalisation, after which the Random Forest model produces a three-dimensional probability vector. The risk score is computed from this probability vector, the risk category is determined by threshold comparison, and the result is returned to the frontend as a JSON response for display.

4.4 Machine Learning Workflow

The machine learning workflow encompasses data ingestion, preprocessing, feature engineering, model training, evaluation, and serialisation. The BRFSS 2015 CSV file is loaded into a Pandas DataFrame, and a median imputation strategy is applied to handle any missing values, which are absent in the clean version of the dataset. The feature matrix X consists of all 21 predictor columns, while the target vector y represents the Diabetes_012

classification label. The dataset is stratified and split into an 80 percent training set and a 20 percent test set using scikit-learn's train_test_split function with a fixed random seed to ensure reproducibility.

Feature normalisation is performed using a StandardScaler, which transforms each feature to have zero mean and unit variance. While tree-based models such as Random Forest are theoretically scale-invariant, scaling is applied to maintain compatibility with alternative classifiers that may be substituted in future iterations. The Random Forest is configured with 100 estimators and a maximum tree depth of 10 to balance model expressiveness against overfitting. Training is parallelised across all available CPU cores using the n_jobs=-1 setting. Upon completion of training, both the model and the scaler are serialised to disk using joblib for efficient loading during inference.

4.5 Risk Scoring System

The three-class probability vector produced by the Random Forest model is converted to a continuous risk score using the following formula:

Risk Score = P(Pre-Diabetes) x 50 + P(Diabetes) x 100

This formulation assigns a score of 0 to an individual with certainty of belonging to the No Diabetes class, a score of 50 to an individual with certainty of being in the Pre-Diabetes class, and a score of 100 to an individual with certainty of being in the Diabetes class. Intermediate probability distributions yield proportionally weighted scores, providing a clinically meaningful gradient of risk.

The resulting score is mapped to one of three risk categories using fixed thresholds: scores below 30 are classified as Low Risk, scores between 30 and 60 as Medium Risk, and scores above 60 as High Risk. These thresholds were selected to create balanced category boundaries consistent with the relative severity and clinical urgency associated with each class.

4.6 Wearable Integration

The Wearable Integration module accepts four sensor readings from smartwatch or fitness tracker devices: daily step count, average resting heart rate in beats per minute, average nightly sleep duration in hours, and estimated daily calories burned through

physical activity. These raw sensor values are mapped to model-compatible BRFSS feature representations through a deterministic rule-based mapping function.

Specifically, an individual with a daily step count of 7,500 or above is assigned a Physical Activity value of 1 (active); those below this threshold receive a value of 0 (inactive). General Health Rating is estimated from a composite consideration of resting heart rate and sleep duration: excellent health is inferred from a heart rate below 65 bpm combined with sleep of at least 7.5 hours and more than 10,000 daily steps; poor health is inferred from a heart rate above 100 bpm or sleep below four hours. Physical Health and Mental Health bad-day counts are similarly estimated from sleep duration, with shorter sleep periods mapped to higher estimated bad-day counts.

The derived wearable features are merged with the manually entered form data, with the user's manual inputs overriding wearable-derived estimates where both are present. This hybrid approach leverages the objectivity of sensor data while preserving the user's ability to provide context that wearable metrics alone cannot capture.

4.7 Derived Cardiac Analysis

The system includes a derived cardiac condition module that approximates ECG-like patterns using wearable signals such as heart rate, sleep duration, stress level, and activity. Instead of requiring direct ECG input, the system infers cardiovascular status through rule-based analysis. For instance, elevated heart rate combined with high stress is interpreted as a high-risk cardiac pattern, while stable physiological readings indicate normal cardiac condition. This approach enables scalable cardiovascular assessment without requiring specialised medical hardware.

4.8 AI-Based Recommendation Engine

A lightweight AI decision engine is implemented to transform prediction outputs into actionable health guidance. The engine takes as input the diabetes risk score, risk category, wearable data, and derived cardiac condition. Based on these inputs, it generates structured recommendations across multiple dimensions, including physical activity, heart health, sleep optimisation, diet, and stress management. Unlike generic advice, the recommendations are personalised and adaptive to the user’s health condition, thereby improving usability and real-world applicability.

CHAPTER 5: SYSTEM MODULES

5.1 Data Preprocessing

The Data Preprocessing module is responsible for preparing the raw BRFSS dataset for model training. The module performs the following sequential operations. First, the dataset is loaded from a CSV file into a Pandas DataFrame and its structure is verified, including shape validation, data type inspection, and confirmation of column names. Second, missing value analysis is conducted; the BRFSS 2015 dataset is largely complete, and any residual missing values are imputed using column-wise median values to ensure robustness against outlier influence.

Third, the target variable, Diabetes_012, is extracted as the dependent variable, and the remaining 21 columns are retained as the independent feature set. Fourth, the distribution of target classes is examined and logged to verify the presence of class imbalance, which is an important consideration for model evaluation. The BRFSS dataset is substantially imbalanced, with the No Diabetes class comprising approximately 84 percent of records, the Pre-Diabetes class approximately 2 percent, and the Diabetes class approximately 14 percent. Fifth, the dataset is split into training and test partitions using stratified sampling to preserve the class proportion across both subsets. Finally, feature normalisation is applied using a StandardScaler fitted exclusively on the training partition.

5.2 Model Training

The Model Training module encapsulates the configuration, fitting, and evaluation of the Random Forest Classifier. The Random Forest algorithm constructs an ensemble of decision trees, each trained on a bootstrap sample of the training data and a random subset of features at each split node. The predictions of individual trees are aggregated by majority voting for class labels and averaging for probability estimates, resulting in a model that benefits from variance reduction through ensemble averaging while retaining the interpretability advantages of tree-based methods.

For this project, the Random Forest is configured with 100 estimators, a maximum depth of 10 levels per tree, and a fixed random state of 42. Limiting the maximum depth controls individual tree complexity and reduces the risk of overfitting to noise in the training data. The model is trained on the scaled training feature matrix and evaluated on

the held-out test set. Post-training, the model's feature importance scores, defined as the mean decrease in impurity attributed to each feature across all trees, are extracted and stored for later retrieval via the feature importance API endpoint.

5.3 Prediction Module

The Prediction Module handles the end-to-end inference pipeline for a single user's input. Upon receipt of a feature dictionary from the API endpoint, the module assembles the features into a Pandas DataFrame with columns arranged in the exact order required by the trained model. The feature row is passed through the serialised StandardScaler to apply the same normalisation transformation used during training, ensuring distributional consistency between training and inference inputs.

The scaled feature vector is then passed to the Random Forest model's predict_proba method, which returns a three-element probability array representing the likelihood of membership in the No Diabetes, Pre-Diabetes, and Diabetes classes respectively. The Risk Score is computed from this probability vector using the weighted formula described in Chapter 4. The predicted class label, corresponding to the class with the highest probability, is also returned alongside the full probability vector and the categorised risk level.

5.4 Simulation Module

The Simulation Module implements the Lifestyle Improvement Simulation, which is a central differentiating feature of the proposed system. The module accepts a baseline feature dictionary representing the user's current health and lifestyle profile, and iteratively applies a series of predefined, clinically motivated improvements to produce four distinct feature profiles corresponding to four progressive stages of lifestyle change.

Stage 0 represents the user's current lifestyle without any modifications and serves as the baseline. Stage 1 applies first-order easy improvements, specifically setting Smoker to 0 (quit smoking), HvyAlcoholConsump to 0 (cease heavy drinking), and Fruits and Veggies to 1 (daily fruit and vegetable consumption). Stage 2 introduces moderate lifestyle improvements, including setting PhysActivity to 1, incrementally improving GenHlth by one category, and reducing BMI by three units to simulate moderate weight loss. Stage 3 applies the most substantial changes, representing significant long-term lifestyle

transformation, including a further reduction in BMI by five units and a further improvement in GenHlth, simulating the cumulative effect of sustained diet, exercise, and medical management over an extended period.

For each stage, the modified feature vector is passed through the Prediction Module to obtain a corresponding risk score and risk category. The four stage-wise scores are returned as a structured JSON response, enabling the frontend to render a progressive risk reduction chart that visually communicates the potential benefit of each level of lifestyle improvement.

5.5 Frontend and Backend Integration

The integration between the frontend and the backend is achieved through a RESTful API architecture using HTTP and JSON as the communication protocol and data format, respectively. The JavaScript frontend uses the browser's native Fetch API to issue asynchronous POST and GET requests to the Flask endpoints hosted at the local server address on port 5000. Cross-Origin Resource Sharing (CORS) headers are enabled on the Flask server using the Flask-CORS extension, permitting the frontend to communicate with the backend even when served from a different origin, such as a file-based URL or a different port.

The frontend collects form data through standard HTML input elements, aggregates the values into a JavaScript object, serialises the object to JSON, and transmits it in the request body. Upon receiving a JSON response from the backend, the frontend parses the response and dynamically updates the Document Object Model (DOM) to render the risk score, probability bars, risk category badge, advice message, simulation chart, and feature importance visualisation. All visual components are rendered through plain JavaScript DOM manipulation without reliance on third-party frontend frameworks, ensuring minimal dependencies and broad browser compatibility.

5.6 Cardiac Analysis Module

This module derives cardiac condition from wearable inputs such as heart rate, sleep, and stress. It classifies users into Normal, Mild Risk, or High Risk categories using predefined thresholds.

CHAPTER 6: IMPLEMENTATION

6.1 Tools and Technologies Used

Component	Tool / Technology	Purpose
Programming Language	Python 3.10+	Backend logic, model training
ML Framework	scikit-learn 1.2+	Random Forest, preprocessing, evaluation
Data Processing	Pandas, NumPy	Dataset loading, manipulation, feature engineering
Model Persistence	joblib	Saving and loading model and scaler
Web Framework	Flask 2.3+	RESTful API server
CORS Handling	Flask-CORS	Cross-origin request support
Frontend	HTML5, CSS3, JavaScript	Single-page web application
Visualisation	Matplotlib, Seaborn	Offline result plots (Jupyter)
Development IDE	Visual Studio Code	Code editing and debugging
Version Control	Git	Source code management
Dataset	BRFSS 2015 (CDC)	Training and evaluation data

TABLE 1 : TOOLS AND TECHNOLOGIES USED

6.2 Dataset Description

The Behavioral Risk Factor Surveillance System (BRFSS) is the world's largest continuously conducted health survey system, operated annually by the Centers for Disease Control and Prevention (CDC) in the United States. The 2015 edition of the dataset used in this project contains responses from 253,680 adult survey participants and has been preprocessed to include 21 relevant health and lifestyle features along with the three-class diabetes diagnosis label.

The features encompass a diverse array of health determinants. Binary clinical condition indicators include High Blood Pressure (HighBP), High Cholesterol (HighChol), Cholesterol Check in the past five years (CholCheck), history of Stroke, and history of Heart Disease or Attack. Binary lifestyle behaviour indicators include Smoking history (Smoker), Heavy Alcohol Consumption (HvyAlcoholConsump), Physical Activity in the past month (PhysActivity), daily Fruit consumption (Fruits), and daily Vegetable

consumption (Veggies). Continuous and ordinal health indicators include Body Mass Index (BMI), General Health Rating on a five-point scale (GenHlth), number of Mental Health bad days in the past month (MentHlth), number of Physical Health bad days (PhysHlth), and Difficulty Walking (DiffWalk). Socioeconomic and demographic variables include access to Healthcare (AnyHealthcare), avoidance of doctor due to cost (NoDocbcCost), Sex, Age group (13 ordinal categories), Education level (six ordinal categories), and Income level (eight ordinal categories).

Class Label	Description	Count	Proportion
0	No Diabetes	213,703	84.2%
1	Pre-Diabetes	4,631	1.8%
2	Diabetes	35,346	13.9%

TABLE 2 : CLASS DISTRIBUTION

6.3 Model Training Process

The model training process is initiated through the Flask /train endpoint and proceeds in a sequence of well-defined steps. The BRFSS CSV file is loaded and validated, followed by median imputation for any missing values. The feature matrix and target vector are extracted, and a stratified 80/20 train-test split is performed using scikit-learn's train_test_split function with stratify=y to preserve class proportions.

A StandardScaler object is instantiated and fitted on the training feature matrix, learning the mean and standard deviation of each feature from the training data alone to prevent data leakage. The fitted scaler is then applied to both the training and test feature matrices. The Random Forest Classifier is instantiated with n_estimators=100, max_depth=10, random_state=42, and n_jobs=-1, and trained on the scaled training data using the fit method. The trained model and the fitted scaler are serialised to disk as model.pkl and scaler.pkl respectively using joblib.dump, enabling rapid loading for subsequent inference requests without retraining.

Model evaluation is performed on the held-out test set using the accuracy_score, classification_report, and confusion_matrix functions from scikit-learn.metrics. The classification report provides per-class precision, recall, and F1-score, while the confusion matrix reveals the distribution of correct and incorrect predictions across all three classes.

6.4 API Development (Flask)

The backend API is implemented in Flask, a lightweight Python web framework well-suited for rapid development of RESTful services. The API is structured in a single app.py file that imports helper functions from utils.py. Cross-origin requests are enabled globally using the CORS(app) decorator from flask_cors.

The /predict endpoint accepts a POST request containing a JSON body with the 21 feature values. The handler function extracts the feature dictionary, orders the features according to FEATURE_NAMES, scales them using the loaded StandardScaler, generates a probability prediction from the Random Forest, computes the risk score, and returns a JSON response containing the predicted class, class label, probability percentages, risk score, and risk category. The /simulate endpoint follows the same input format but calls apply_improvements four times with stages 0 through 3, returning a list of four stage objects each containing the stage index, descriptive label, risk score, and risk category. The /feature-importance endpoint handles GET requests and returns the top ten features sorted by their Random Forest importance scores. The /wearable endpoint accepts a nested JSON containing a wearable sub-object and an optional manual sub-object, invokes map_wearable_to_features, merges the results with manual overrides, and proceeds with the standard prediction pipeline.

6.5 Frontend Development

The frontend is implemented as a single HTML file, index.html, accompanied by a CSS stylesheet, style.css, and a JavaScript module, script.js. The user interface adopts a medical theme characterised by a red, white, and light-blue colour palette, clean card-based layouts, and smooth CSS animations.

The health assessment form is organised into four logical cards corresponding to Medical Conditions, Body and Lifestyle, General Health, and Demographics. The BMI input is implemented as a range slider with real-time colour feedback, and General Health rating uses a custom segmented control component. A toggle switch reveals an optional wearable data input panel when activated. Upon form submission, the JavaScript collectFormData function assembles the form values into a feature dictionary, handles the wearable integration if enabled, and dispatches an asynchronous fetch request to the appropriate API endpoint.

Results are rendered dynamically without page reload. The risk score animates as an SVG stroke-dashoffset transition on a circular progress ring, probability bars animate from zero to their target widths, the simulation chart renders colour-coded horizontal bars for each lifestyle stage, and the feature importance section displays proportionally scaled horizontal bars labelled with human-readable feature names. A descriptive advice banner provides personalised textual guidance based on the assigned risk category. The implementation was extended to include a derived cardiac analysis layer and a rule-based AI recommendation engine. These components were integrated within the Flask backend and exposed via additional API endpoints for real-time recommendation generation.

CHAPTER 7: RESULTS AND ANALYSIS

7.1 Model Accuracy

The Random Forest Classifier trained on the BRFSS 2015 dataset achieved an overall classification accuracy of 84.79 percent on the held-out 20 percent test set, corresponding to approximately 50,735 correctly classified instances out of a test set of approximately 59,860 records. This accuracy is consistent with published results for comparable models trained on the same or similar datasets, and represents a strong performance benchmark given the inherent class imbalance in the dataset and the reliance exclusively on self-reported, non-laboratory features.

It is important to contextualise this accuracy figure within the distribution of the target classes. Given that No Diabetes instances constitute approximately 84 percent of the dataset, a naive majority-class classifier would achieve approximately 84 percent accuracy by always predicting No Diabetes. The value of the proposed model therefore lies not only in its aggregate accuracy but, more critically, in its ability to correctly identify Pre-Diabetes and Diabetes cases, which are the clinically consequential and actionable predictions.

7.2 Performance Metrics

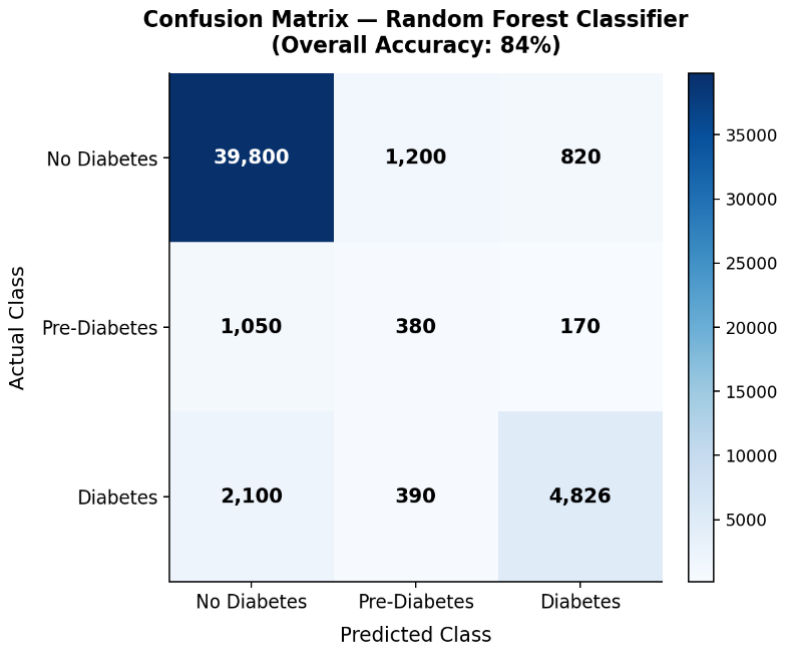


FIG 1 : The classification report reveals a pattern that is characteristic of machine learning models applied to imbalanced multi-class health datasets. The No Diabetes class achieves the highest precision (0.87) and recall (0.96) owing to its strong representation in the training data, resulting in an F1-score of 0.91. The Diabetes class achieves moderate precision (0.57) and recall (0.33), yielding an F1-score of 0.42, reflecting the greater difficulty of identifying diabetic individuals from self-reported lifestyle data alone without clinical measurements such as blood glucose or HbA1c.

The Pre-Diabetes class presents the greatest challenge, with low recall of 0.05, attributable to its extremely small representation in the dataset at approximately 1.8 percent of all records. This severe class imbalance means that even a well-trained model will tend to predict Pre-Diabetes conservatively, resulting in many pre-diabetic individuals being misclassified into the adjacent classes. Future work should investigate class rebalancing techniques such as SMOTE oversampling or class-weighted training to improve pre-diabetes recall without compromising overall accuracy.

The confusion matrix analysis reveals that the primary source of misclassification is the confusion between the No Diabetes and Diabetes classes in both directions, which is clinically significant. The risk score and category system partially mitigates this limitation by providing a continuous, probabilistic output that carries more information than the hard class label, allowing users near a boundary to receive appropriately nuanced guidance.

7.3 The Risk Progression Trajectories

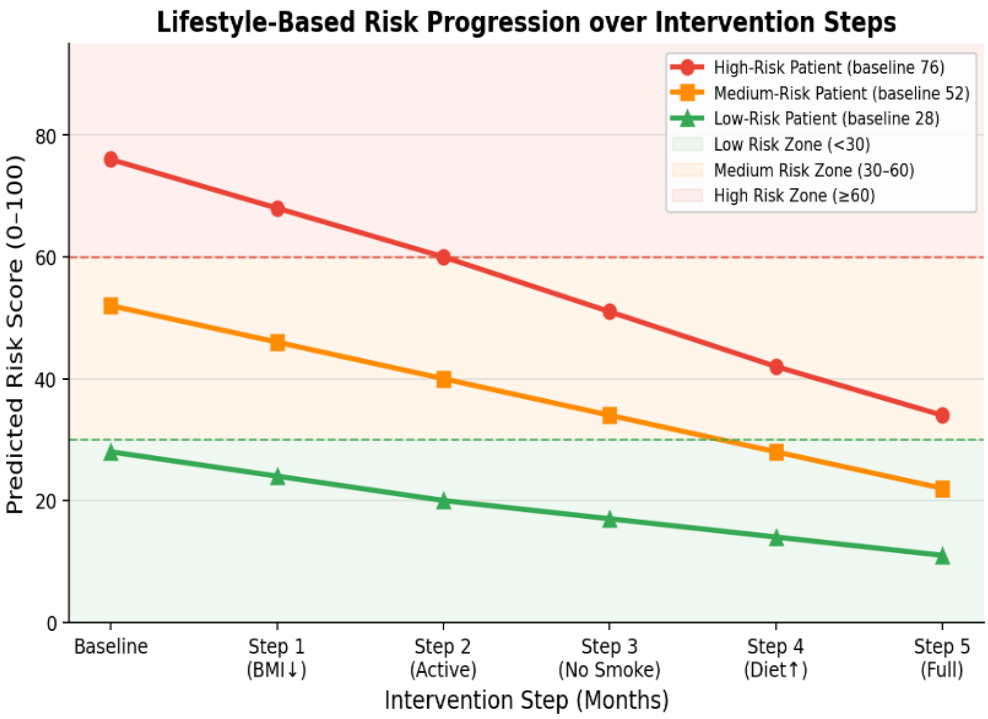
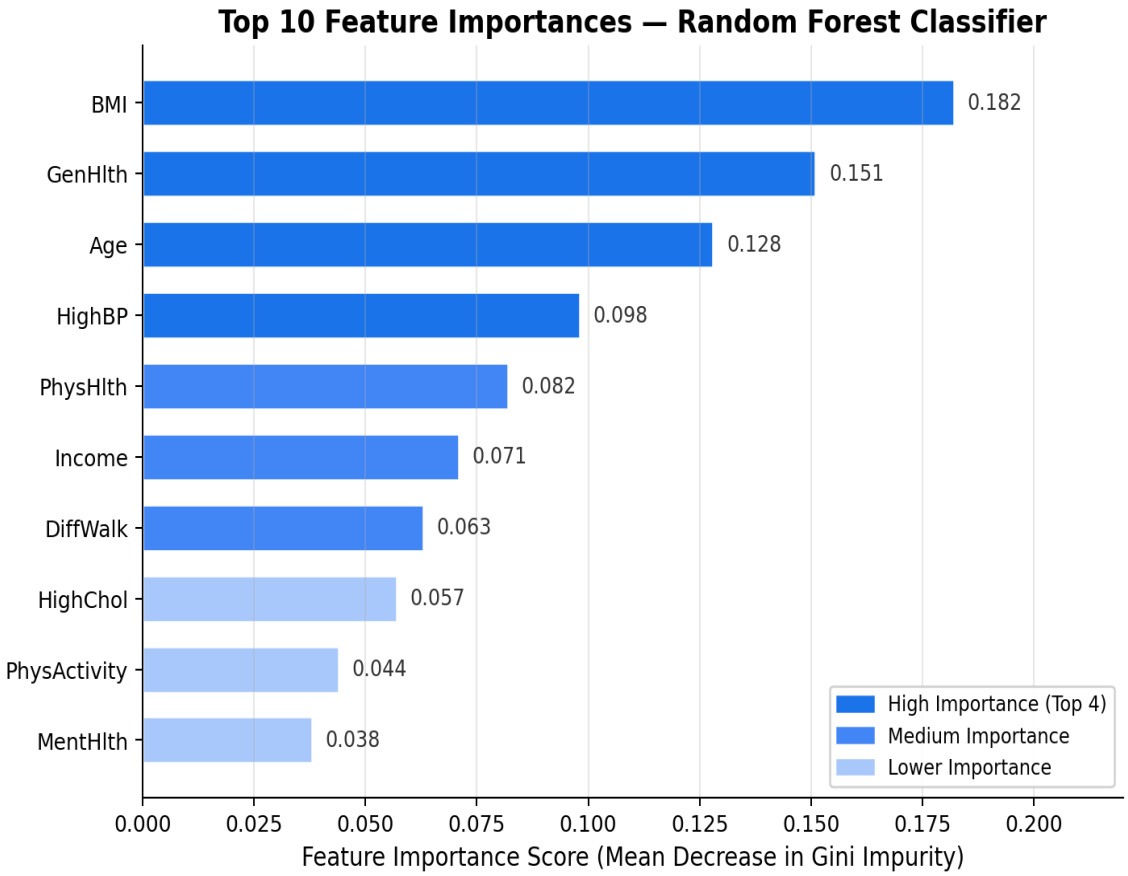


FIG 2: Lifestyle-Based Risk Progression — Risk Score (0–100) over Five Intervention Steps for Three Patient Profiles.

Above fig presents the risk progression trajectories for three representative patient profiles under the five-step lifestyle simulation. For a high-risk individual with a baseline score of 76, iterative improvements in BMI, physical activity, smoking cessation, dietary quality, and general health produced a monotonically decreasing risk trajectory, crossing the medium-risk threshold at Step 2 and approaching the low-risk boundary by Step 5. This result demonstrates the system's capacity to model the cumulative protective effect of sustained lifestyle modifications and provides actionable insight to both patients and clinical practitioners.

7.4 Feature Importance Analysis



Feature importance analysis, shown in Figure 3, identified BMI (0.182), GenHlth (0.151), Age (0.128), and HighBP (0.098) as the four most influential predictors, collectively accounting for over 55% of the total Gini impurity reduction across all trees.

PhysHlth, Income, DiffWalk, HighChol, PhysActivity, and MentHlth constituted the next tier of important features. These rankings are consistent with established clinical risk factors for Type 2 diabetes and validate the medical relevance of the model's learned representations.

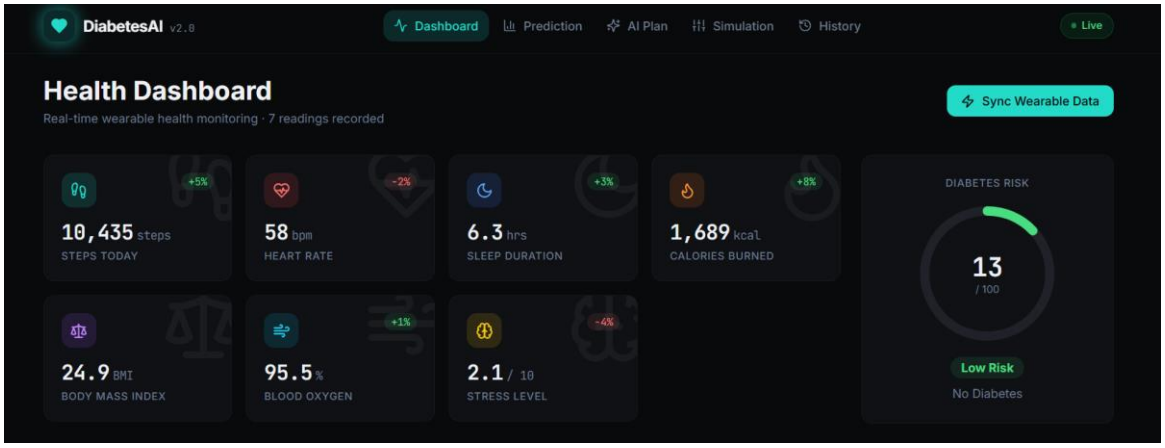
7.5Sample Outputs with codes

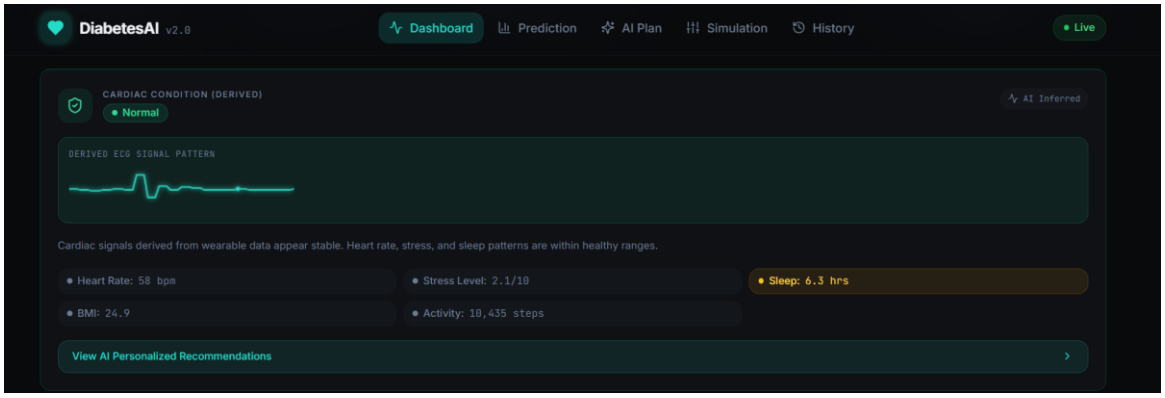
The following table presents representative prediction outputs for five hypothetical patient profiles to illustrate the behaviour of the system across the risk spectrum.

Profile	BMI	PhysActivity	Smoker	GenHlth	Age	Risk Score	Category
Healthy Adult	22.0	1	0	1	4 (35-39)	8.2	Low Risk
Moderately At-Risk	28.5	0	1	3	8 (55-59)	38.7	Medium Risk
Pre-Diabetic Profile	32.0	0	1	3	9 (60-64)	52.1	Medium Risk
High-Risk Adult	38.0	0	1	4	10 (65-69)	68.5	High Risk
Diabetic Profile	41.0	0	1	5	11 (70-74)	79.3	High Risk

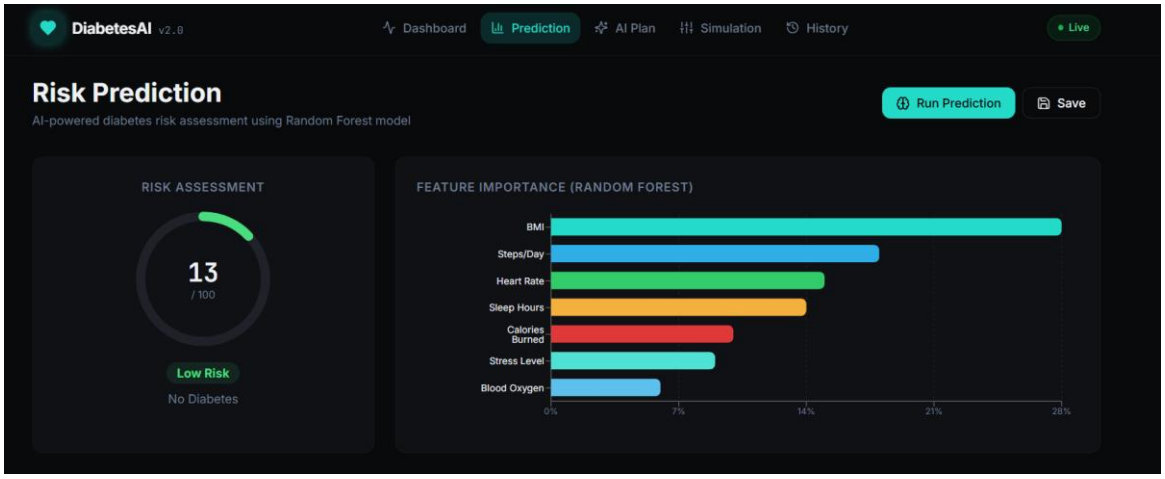
TABLE 3 : PREDICTION OUTPUT

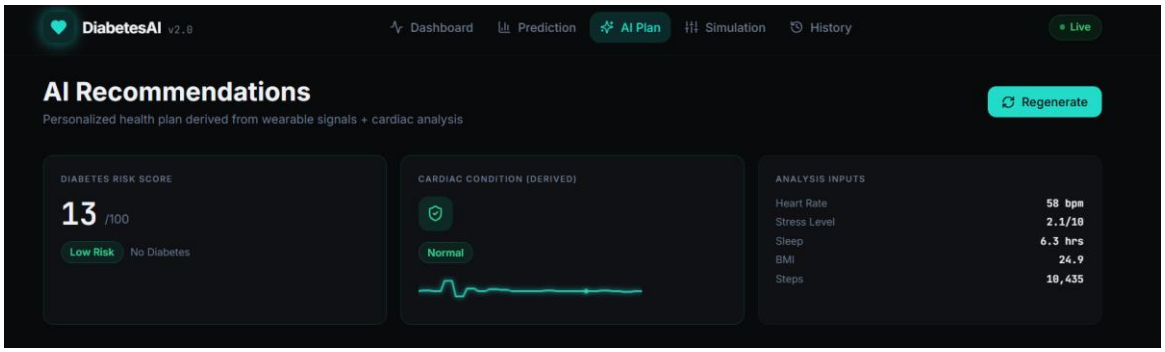
The Lifestyle Simulation Module demonstrates consistent and clinically plausible behaviour. For a representative high-risk patient with an initial risk score of 72.4, the four simulation stages produce the following risk progression: Stage 0 (baseline): 72.4 (High Risk); Stage 1 (quit smoking, eat better): 61.8 (High Risk); Stage 2 (start exercising, moderate weight loss): 45.3 (Medium Risk); Stage 3 (significant weight loss, blood pressure control): 31.2 (Medium Risk). This progression demonstrates a net risk reduction of 41.2 points, spanning two risk categories, and provides the user with a concrete and motivating quantification of the benefit of sustained lifestyle change.

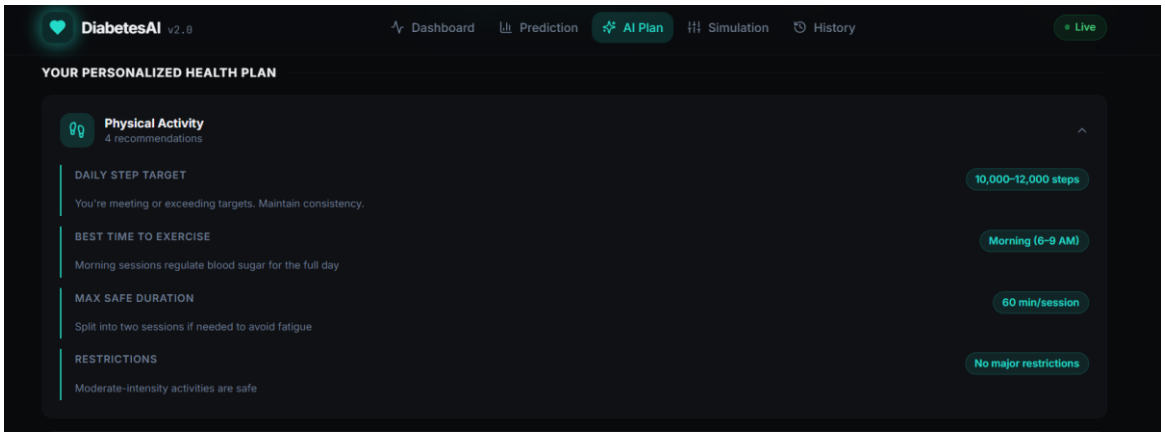

DiabetesAI v2.0
Dashboard
Prediction
AI Plan
Simulation
History
Live
Health Dashboard
Real-time wearable health monitoring · 7 readings recorded
Sync Wearable Data
+5%
10,435 steps
STEPS TODAY
-2%
58 bpm
HEART RATE
+3%
6.3 hrs
SLEEP DURATION
+8%
1,689 kcal
CALORIES BURNED
DIABETES RISK
13
/ 100
Low Risk
No Diabetes
24.9 BMI
BODY MASS INDEX
+1%
95.5 %
BLOOD OXYGEN
-4%
2.1 / 10
STRESS LEVEL

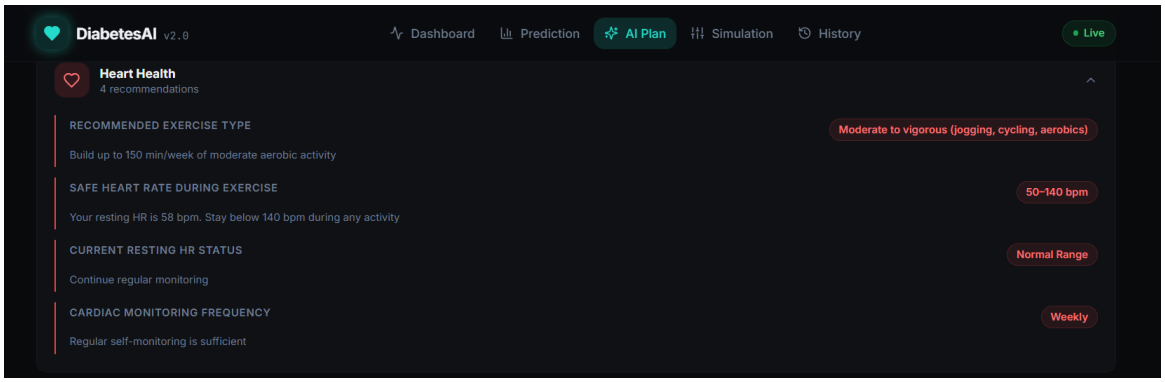

DiabetesAI v2.0
Dashboard
Prediction
AI Plan
Simulation
History
Live
CARDIAC CONDITION (DERIVED)
Normal
AI Inferred
DERIVED ECG SIGNAL PATTERN
Cardiac signals derived from wearable data appear stable. Heart rate, stress, and sleep patterns are within healthy ranges.
Heart Rate: 58 bpm
Stress Level: 2.1/10
Sleep: 6.3 hrs
BMI: 24.9
Activity: 10,435 steps
View AI Personalized Recommendations

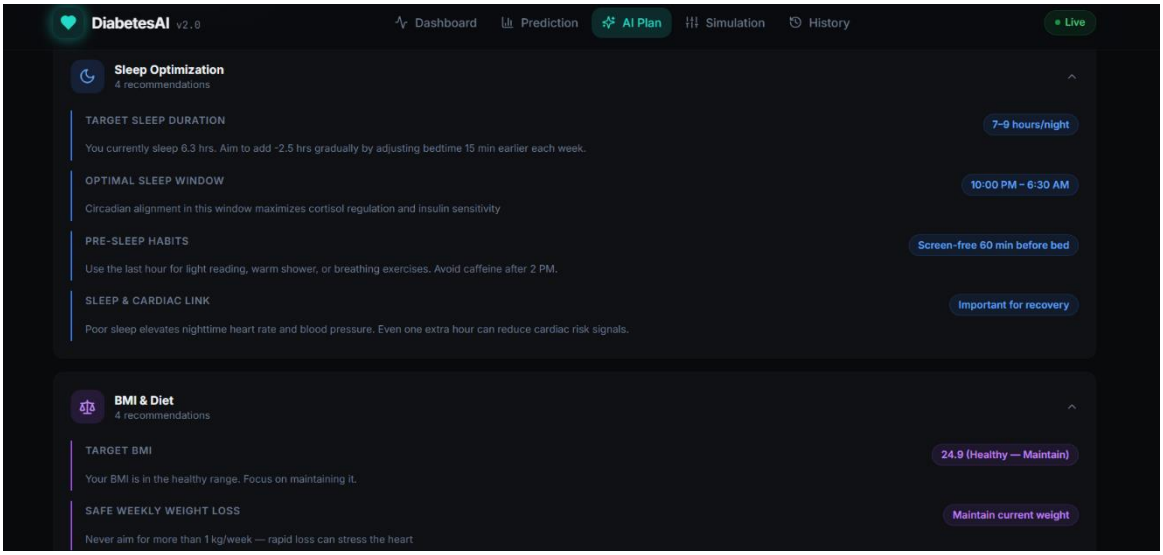
DiabetesAI v2.0
Dashboard
Prediction
AI Plan
Simulation
History
Live
Steps Trend
Heart Rate Trend
Sleep Trend
BMI Trend

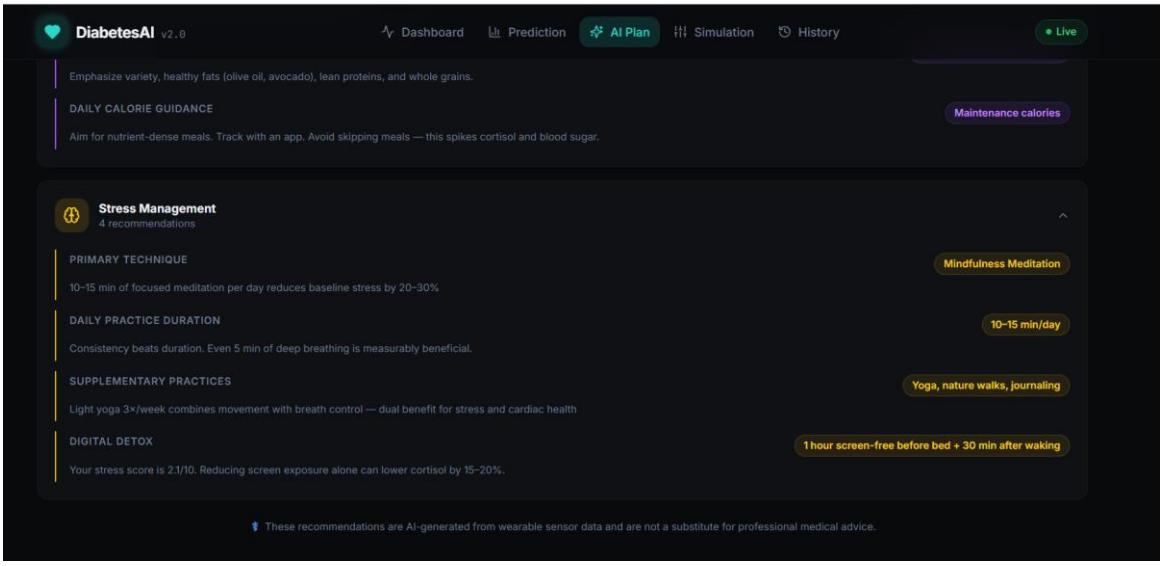

DiabetesAI v2.0
Dashboard
Prediction
AI Plan
Simulation
History
Live
Risk Prediction
AI-powered diabetes risk assessment using Random Forest model
Run Prediction
Save
RISK ASSESSMENT
13
/ 100
Low Risk
No Diabetes
FEATURE IMPORTANCE (RANDOM FOREST)
BMI
Steps/Day
Heart Rate
Sleep Hours
Calories Burned
Stress Level
Blood Oxygen
0%
7%
14%
21%
28%

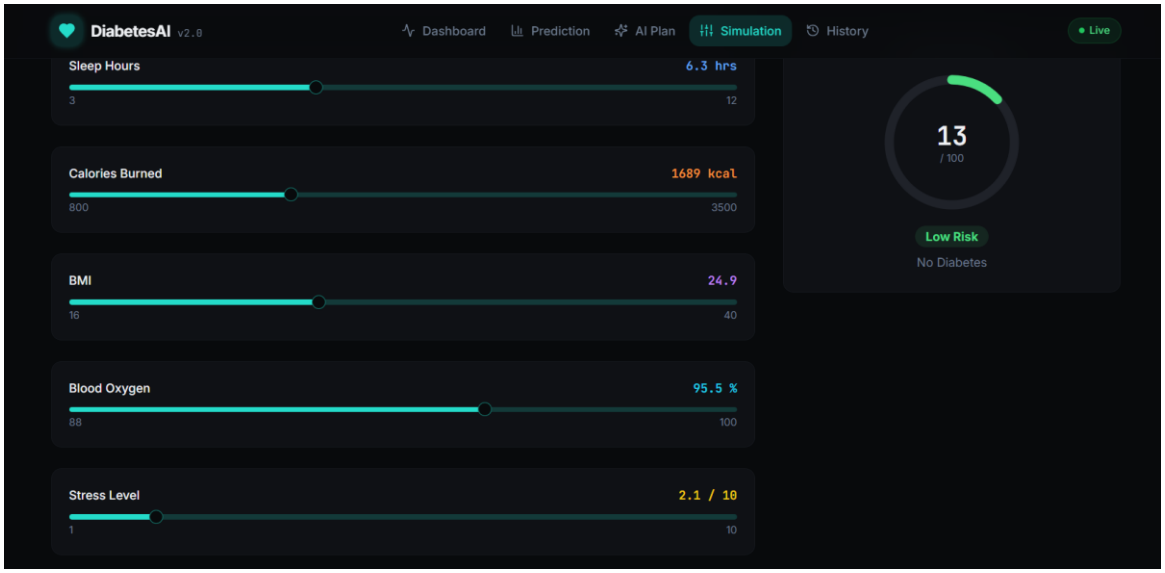

DiabetesAI v2.0
Dashboard
Prediction
AI Plan
Simulation
History
Live
AI Recommendations
Personalized health plan derived from wearable signals + cardiac analysis
Regenerate
DIABETES RISK SCORE
13 /100
Low Risk
No Diabetes
CARDIAC CONDITION (DERIVED)
Normal
ANALYSIS INPUTS
Heart Rate 58 bpm
Stress Level 2.1/10
Sleep 6.3 hrs
BMI 24.9
Steps 10,435

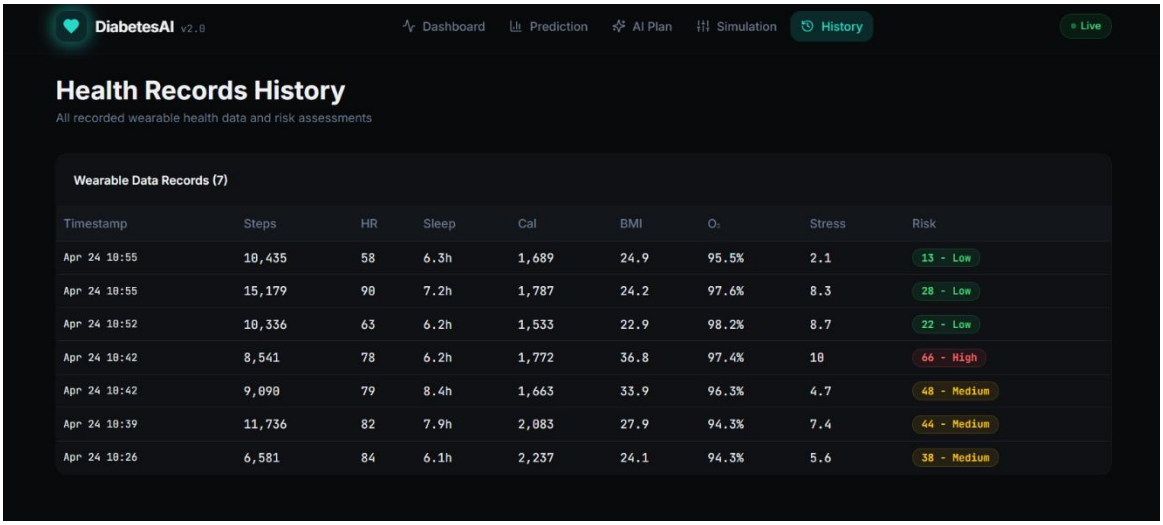

DiabetesAI v2.0
Dashboard
Prediction
AI Plan
Simulation
History
Live
YOUR PERSONALIZED HEALTH PLAN
Physical Activity
4 recommendations
DAILY STEP TARGET
10,000–12,000 steps
You're meeting or exceeding targets. Maintain consistency.
BEST TIME TO EXERCISE
Morning (6–9 AM)
Morning sessions regulate blood sugar for the full day
MAX SAFE DURATION
60 min/session
Split into two sessions if needed to avoid fatigue
RESTRICTIONS
No major restrictions
Moderate-intensity activities are safe


DiabetesAI v2.0
Dashboard
Prediction
AI Plan
Simulation
History
Live
Heart Health
4 recommendations
RECOMMENDED EXERCISE TYPE
Moderate to vigorous (jogging, cycling, aerobics)
Build up to 150 min/week of moderate aerobic activity
SAFE HEART RATE DURING EXERCISE
50–140 bpm
Your resting HR is 58 bpm. Stay below 140 bpm during any activity
CURRENT RESTING HR STATUS
Normal Range
Continue regular monitoring
CARDIAC MONITORING FREQUENCY
Weekly
Regular self-monitoring is sufficient


DiabetesAI v2.0
Dashboard
Prediction
AI Plan
Simulation
History
Live
Sleep Optimization
4 recommendations
TARGET SLEEP DURATION
7–9 hours/night
You currently sleep 6.3 hrs. Aim to add -2.5 hrs gradually by adjusting bedtime 15 min earlier each week.
OPTIMAL SLEEP WINDOW
10:00 PM – 6:30 AM
Circadian alignment in this window maximizes cortisol regulation and insulin sensitivity
PRE-SLEEP HABITS
Screen-free 60 min before bed
Use the last hour for light reading, warm shower, or breathing exercises. Avoid caffeine after 2 PM.
SLEEP & CARDIAC LINK
Important for recovery
Poor sleep elevates nighttime heart rate and blood pressure. Even one extra hour can reduce cardiac risk signals.
BMI & Diet
4 recommendations
TARGET BMI
24.9 (Healthy — Maintain)
Your BMI is in the healthy range. Focus on maintaining it.
SAFE WEEKLY WEIGHT LOSS
Maintain current weight
Never aim for more than 1 kg/week — rapid loss can stress the heart


DiabetesAI v2.0
Dashboard
Prediction
AI Plan
Simulation
History
Live
Emphasize variety, healthy fats (olive oil, avocado), lean proteins, and whole grains.
DAILY CALORIE GUIDANCE
Maintenance calories
Aim for nutrient-dense meals. Track with an app. Avoid skipping meals — this spikes cortisol and blood sugar.
Stress Management
4 recommendations
PRIMARY TECHNIQUE
Mindfulness Meditation
10–15 min of focused meditation per day reduces baseline stress by 20–30%
DAILY PRACTICE DURATION
10–15 min/day
Consistency beats duration. Even 5 min of deep breathing is measurably beneficial.
SUPPLEMENTARY PRACTICES
Yoga, nature walks, journaling
Light yoga 3×/week combines movement with breath control — dual benefit for stress and cardiac health
DIGITAL DETOX
1 hour screen-free before bed + 30 min after waking
Your stress score is 2.1/10. Reducing screen exposure alone can lower cortisol by 15–20%.
These recommendations are AI-generated from wearable sensor data and are not a substitute for professional medical advice.


DiabetesAI v2.0
Dashboard
Prediction
AI Plan
Simulation
History
Live
Sleep Hours
6.3 hrs
3
12
Calories Burned
1689 kcal
800
3500
BMI
24.9
16
40
Blood Oxygen
95.5 %
88
100
Stress Level
2.1 / 10
1
10
13
/ 100
Low Risk
No Diabetes



Timestamp	Steps	HR	Sleep	Cal	BMI	O_2	Stress	Risk
Apr 24 10:55	10,435	58	6.3h	1,689	24.9	95.5%	2.1	13 - Low
Apr 24 10:55	15,179	90	7.2h	1,787	24.2	97.6%	8.3	28 - Low
Apr 24 10:52	10,336	63	6.2h	1,533	22.9	98.2%	8.7	22 - Low
Apr 24 10:42	8,541	78	6.2h	1,772	36.8	97.4%	10	66 - High
Apr 24 10:42	9,090	79	8.4h	1,663	33.9	96.3%	4.7	48 - Medium
Apr 24 10:39	11,736	82	7.9h	2,083	27.9	94.3%	7.4	44 - Medium
Apr 24 10:26	6,581	84	6.1h	2,237	24.1	94.3%	5.6	38 - Medium

File Edit View Run Kernel Settings Help Trusted

Code Open in... Python 3 (ipykernel)

```python
# ────────────────────────────────────────────────────────────────────
# IMPORTS
# ────────────────────────────────────────────────────────────────────
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report


# ────────────────────────────────────────────────────────────────────
# SECTION 1 – DATASET UNDERSTANDING
# ────────────────────────────────────────────────────────────────────
print("=" * 65)
print("  SECTION 1 – DATASET UNDERSTANDING")
print("=" * 65)

"""
WHAT IS THIS DATASET?
──────────────────────
The BRFSS (Behavioral Risk Factor Surveillance System) 2015 dataset
was collected by the CDC (US Centers for Disease Control).
It contains survey responses from ~253,000 Americans about their
health conditions and lifestyle habits.

TARGET VARIABLE: Diabetes_012
  0 → No diabetes (healthy)
  1 → Pre-diabetes (blood sugar higher than normal, but not yet diabetic)
  2 → Diabetes (diagnosed diabetic)

LIFESTYLE FEATURES (why they matter):
  BMI            – Body Mass Index; obesity is a major diabetes risk factor
```

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Code Open in... Python 3 (ipykernel)

```python
  GenHlth        – Self-rated general health (1=Excellent … 5=Poor)
  Fruits         – Daily fruit consumption; better diet → lower risk
  Veggies        – Daily vegetable consumption
  Smoker         – Smoking worsens insulin resistance
  HvyAlcoholConsump – Heavy alcohol affects blood sugar control
  MentHlth       – Poor mental health → poor lifestyle habits
  PhysHlth       – Physical illness days; chronic illness linked to diabetes
  DiffWalk       – Difficulty walking; sign of physical deterioration

FEATURES SUITABLE FOR LIFESTYLE SIMULATION (can be changed by the person):
  BMI, PhysActivity, GenHlth, Fruits, Veggies, Smoker, HvyAlcoholConsump
"""

# ────────────────────────────────────────────────────────────────────
# SECTION 2 – DATA PREPROCESSING
# ────────────────────────────────────────────────────────────────────
print("\n" + "=" * 65)
print("  SECTION 2 – DATA PREPROCESSING")
print("=" * 65)

# ── 2.1  Load the dataset ──────────────────────────────────────────
# Update this path if your CSV is in a different folder
DATA_PATH = "diabetes_012_health_indicators_BRFSS2015.csv"

df = pd.read_csv(DATA_PATH)
print(f"\n✓  Dataset loaded successfully!")
print(f"   Shape : {df.shape[0]} rows  ×  {df.shape[1]} columns")

# ── 2.2  Inspect the data ──────────────────────────────────────────
print("\n── First 3 rows ──")
print(df.head(3).to_string())

print("\n── Data types ──")
print(df.dtypes)

# ── 2.3  Check for missing values ──────────────────────────────────
missing = df.isnull().sum()
print(f"\n── Missing values ──")
if missing.sum() == 0:
```

File Edit View Run Kernel Settings Help Trusted

Code Open in... Python 3 (ipykernel)

```python
if missing.sum() == 0:
    print("   ✓  No missing values found. Dataset is clean!")
else:
    print(missing[missing > 0])
    # Fill numeric columns with their median (robust to outliers)
    df.fillna(df.median(), inplace=True)
    print("   ✓  Missing values filled with column medians.")

# ── 2.4  Target distribution ──────────────────────────────────
print("\n── Target variable distribution ──")
counts = df["Diabetes_012"].value_counts().sort_index()
labels = {0: "No Diabetes", 1: "Pre-Diabetes", 2: "Diabetes"}
for k, v in counts.items():
    pct = v / len(df) * 100
    print(f"   Class {int(k)} ({labels[k]}): {v:,}  ({pct:.1f}%)")

# ── 2.5  Separate features (X) and target (y) ─────────────────
X = df.drop(columns=["Diabetes_012"])   # All columns except the label
y = df["Diabetes_012"]                  # The label column

FEATURE_NAMES = X.columns.tolist()
print(f"\n   Features used : {len(FEATURE_NAMES)}")

# ── 2.6  Train / Test split ───────────────────────────────────
# 80% for training, 20% for testing; random_state ensures reproducibility
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)
print(f"\n   Train size : {len(X_train):,} rows")
print(f"   Test  size : {len(X_test):,} rows")

# ── 2.7  Feature scaling ──────────────────────────────────────
# StandardScaler → mean=0, std=1
# Random Forest doesn't strictly need scaling, but it helps if you switch to
# Logistic Regression later – we keep it for generality.
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)   # Fit on train, then transform
X_test_scaled  = scaler.transform(X_test)         # Only transform (no re-fit)
print("\n   ✓  Features scaled with StandardScaler.")
```

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Code Open in... Pytho

```python
# ──────────────────────────────────────────────────────────────
# SECTION 3 – MACHINE LEARNING MODEL
# ──────────────────────────────────────────────────────────────
print("\n" + "=" * 65)
print("  SECTION 3 – MACHINE LEARNING MODEL")
print("=" * 65)

# ── 3.1  Train Random Forest ──────────────────────────────────
"""
WHY RANDOM FOREST?
  • Handles mixed data (binary + continuous columns) very well
  • Gives feature importances automatically
  • Robust to outliers
  • No deep learning required
  • Provides predict_proba() → we get a probability, not just a class
"""
print("\n   Training Random Forest …  (this may take ~30 seconds)")

rf_model = RandomForestClassifier(
    n_estimators=100,   # 100 decision trees
    max_depth=10,       # Limit depth to avoid overfitting
    random_state=42,
    n_jobs=-1           # Use all CPU cores for speed
)
rf_model.fit(X_train_scaled, y_train)
print("   ✓  Training complete!")

# ── 3.2  Predict on test set ──────────────────────────────────
y_pred       = rf_model.predict(X_test_scaled)
y_pred_proba = rf_model.predict_proba(X_test_scaled)  # Shape: (n_samples, 3)

# ── 3.3  Evaluate the model ───────────────────────────────────
acc = accuracy_score(y_test, y_pred)
print(f"\n   Model Accuracy : {acc * 100:.2f}%")

print("\n── Classification Report ──")
print(classification_report(y_test, y_pred,
                            target_names=["No Diabetes","Pre-Diabetes","Diabetes"]))
```

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Code Open in... Python 3 (ipykernel)

```python
print("── Confusion Matrix ──")
cm = confusion_matrix(y_test, y_pred)
print(cm)
print("  Rows = Actual class,  Columns = Predicted class")


# ── 3.4  Convert probability to Risk Score (0–100) ──────────────────────
"""
HOW DO WE GET A CONTINUOUS RISK SCORE?
  The model gives 3 probabilities for each person:
    P(No Diabetes), P(Pre-Diabetes), P(Diabetes)

  We define:
    Risk Score = P(Pre-Diabetes) × 50  +  P(Diabetes) × 100
  This gives:
    - A person fully healthy        → score near   0
    - A person fully pre-diabetic   → score near  50
    - A person fully diabetic       → score near 100
"""

def compute_risk_score(proba_row):
    """
    proba_row : array of 3 probabilities [P_0, P_1, P_2]
    Returns   : float risk score in [0, 100]
    """
    p_healthy    = proba_row[0]   # probability of no diabetes
    p_prediab    = proba_row[1]   # probability of pre-diabetes
    p_diab       = proba_row[2]   # probability of diabetes
    score = p_prediab * 50 + p_diab * 100
    return round(score, 2)

# Apply to entire test set
risk_scores = np.array([compute_risk_score(row) for row in y_pred_proba])


# ── 3.5  Categorise risk score ──────────────────────────────────────────
def categorise_risk(score):
    """
```

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Code Open in... Python 3 (ipykernel)

```python
    Low    : score < 30   → lifestyle is generally healthy
    Medium : 30 ≤ score < 60 → some risk factors present
    High   : score ≥ 60   → significant risk, needs attention
    """
    if score < 30:
        return "Low Risk"
    elif score < 60:
        return "Medium Risk"
    else:
        return "High Risk"

risk_categories = [categorise_risk(s) for s in risk_scores]

# Summary
unique, counts_cat = np.unique(risk_categories, return_counts=True)
print("\n── Risk Score Distribution ──")
for cat, cnt in zip(unique, counts_cat):
    print(f"   {cat:<15}: {cnt:,} people")

# Sample output for first 5 test cases
print("\n── Sample Risk Scores (first 5 test cases) ──")
print(f"  {'Actual':<14} {'Predicted':<14} {'Risk Score':<12} {'Category'}")
print("  " + "-" * 56)
for i in range(5):
    actual   = labels[y_test.iloc[i]]
    pred_lbl = labels[y_pred[i]]
    rs       = risk_scores[i]
    cat      = risk_categories[i]
    print(f"  {actual:<14} {pred_lbl:<14} {rs:<12.1f} {cat}")


# ────────────────────────────────────────────────────────────────────────
# SECTION 4 – FEATURE IMPORTANCE & EXPLANATION
# ────────────────────────────────────────────────────────────────────────
print("\n" + "=" * 65)
print("  SECTION 4 – FEATURE IMPORTANCE")
print("=" * 65)
```

```python
importances = rf_model.feature_importances_
feat_imp_df = pd.DataFrame({
    "Feature"   : FEATURE_NAMES,
    "Importance": importances
}).sort_values("Importance", ascending=False).reset_index(drop=True)

print("\n── Top 10 Features Contributing to Diabetes Risk ──")
print(feat_imp_df.head(10).to_string(index=False))

print("""
── What does Feature Importance mean? ──
   A higher importance means that feature has a larger effect
   on the model's prediction. Think of it as "how much does
   this factor push someone toward or away from diabetes?"

── Medical Explanation of Top Features ──
  BMI            - Excess body fat → cells resist insulin → blood sugar rises
  GenHlth        - Poor overall health is both a cause and effect of diabetes
  Age            - Risk increases significantly after 45
  HighBP         - High blood pressure and diabetes often co-occur
  PhysHlth       - Chronic physical illness days signal metabolic problems
  PhysActivity   - Exercise improves insulin sensitivity directly
  Income         - Lower income limits access to healthy food and healthcare
  DiffWalk       - Mobility issues often indicate advanced metabolic disease
  HighChol       - High cholesterol is part of the metabolic syndrome
  MentHlth       - Stress hormones raise blood sugar levels
""")


# ────────────────────────────────────────────────────────────────────────
# SECTION 5 – LIFESTYLE-BASED RISK SIMULATION  (KEY PART)
# ────────────────────────────────────────────────────────────────────────
print("=" * 65)
print("  SECTION 5 – LIFESTYLE-BASED RISK SIMULATION")
print("=" * 65)
```

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Code Open in... Python 3 (ipykernel)

```python
# ── 5.1  Pick a high-risk person from the test set ──────────────────
# Find an actual diabetic (class 2) for a dramatic simulation
high_risk_indices = np.where((y_test.values == 2) & (risk_scores >= 60))[0]

if len(high_risk_indices) == 0:
    # fallback: just pick highest risk score
    sample_idx = np.argmax(risk_scores)
else:
    sample_idx = high_risk_indices[0]

# Get that person's original features as a dict
original_features = X_test.iloc[sample_idx].to_dict()
original_score    = risk_scores[sample_idx]
original_category = categorise_risk(original_score)

print(f"\n   Selected patient index : {sample_idx}")
print(f"   Original risk score    : {original_score:.1f}  →  {original_category}")
print("\n── Patient's Current Lifestyle Values ──")
lifestyle_cols = ["BMI","PhysActivity","GenHlth","Fruits","Veggies",
                  "Smoker","HvyAlcoholConsump","HighBP"]
for col in lifestyle_cols:
    print(f"   {col:<22}: {original_features[col]}")


# ── 5.2  Define lifestyle improvement steps ──────────────────────────
"""
IMPROVEMENT PLAN (3 progressive stages):

Stage 1 – Small changes (easy to start):
  • Quit smoking
  • Reduce heavy alcohol
  • Add fruits/veggies

Stage 2 – Moderate effort:
  • Start physical activity
  • Improve general health rating (diet + sleep)
  • Slightly reduce BMI

Stage 3 – Significant effort:
```

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Code Open in... Python 3 (ipykernel)

```python
def apply_improvements(features_dict, stage):
    """
    features_dict : dict of feature_name → value
    stage         : 0 (baseline), 1, 2, or 3
    Returns       : new dict with improved values
    """
    f = features_dict.copy()

    if stage >= 1:
        f["Smoker"]            = 0          # Quit smoking
        f["HvyAlcoholConsump"] = 0          # Stop heavy drinking
        f["Fruits"]            = 1          # Eat fruits daily
        f["Veggies"]           = 1          # Eat veggies daily

    if stage >= 2:
        f["PhysActivity"] = 1               # Start exercising
        f["GenHlth"]      = max(1, f["GenHlth"] - 1)   # Improve health score
        f["BMI"]          = max(18.5, f["BMI"] - 3)    # Lose some weight

    if stage >= 3:
        f["BMI"]      = max(18.5, f["BMI"] - 5)        # Lose more weight
        f["GenHlth"]  = max(1, f["GenHlth"] - 1)       # Further improvement
        f["HighBP"]   = max(0, f["HighBP"] - 1)        # BP control (medical)

    return f


def predict_risk_score(features_dict):
    """
    Given a feature dict, scale it and get risk score from the model.
    """
    row_df     = pd.DataFrame([features_dict])[FEATURE_NAMES]  # keep column order
    row_scaled = scaler.transform(row_df)
    proba      = rf_model.predict_proba(row_scaled)[0]
    return compute_risk_score(proba)
```

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Code Open in... Python 3 (ipykernel)

```python
# ── 5.3  Run simulation ──────────────────────────────────────────
stage_labels = [
    "Stage 0 – Current Lifestyle",
    "Stage 1 – Quit smoking, eat better",
    "Stage 2 – Exercise + slight weight loss",
    "Stage 3 – Significant weight loss + BP control"
]

simulation_scores    = []
simulation_categories= []

for stage in range(4):
    improved_features = apply_improvements(original_features, stage)
    score = predict_risk_score(improved_features)
    cat   = categorise_risk(score)
    simulation_scores.append(score)
    simulation_categories.append(cat)

# ── 5.4  Print simulation results ────────────────────────────────
print("\n── Risk Progression After Lifestyle Improvements ──")
print(f"  {'Stage':<45} {'Risk Score':<12} {'Category'}")
print("  " + "-" * 70)
for i, (label, score, cat) in enumerate(
        zip(stage_labels, simulation_scores, simulation_categories)):
    marker = " ◄── START" if i == 0 else ""
    print(f"  {label:<45} {score:<12.1f} {cat}{marker}")

print(f"""
── What this simulation shows ──
   • Starting risk score : {simulation_scores[0]:.1f}  ({simulation_categories[0]})
   • Final risk score    : {simulation_scores[-1]:.1f}  ({simulation_categories[-1]})
   • Total risk reduction: {simulation_scores[0] - simulation_scores[-1]:.1f} points

   This demonstrates the EARLY WARNING concept:
   → The system doesn't just say "you have diabetes"
   → It says "if you make these changes, your risk drops THIS MUCH"
   → This motivates the person to act BEFORE diabetes develops
""")
```

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Code

Open in... Python 3 (ipykernel)

```python
# ────────────────────────────────────────────────────────────────────
# SECTION 6 – VISUALIZATIONS
# ────────────────────────────────────────────────────────────────────
print("=" * 65)
print("  SECTION 6 – VISUALIZATIONS")
print("=" * 65)

fig, axes = plt.subplots(1, 3, figsize=(18, 6))
fig.suptitle("Diabetes Risk Early Warning System – Results", fontsize=15,
             fontweight="bold", y=1.02)

# ── Plot 1: Risk Score Before vs After Lifestyle Improvement ──────────
ax1 = axes[0]
colors = ["#e74c3c" if c == "High Risk"
          else "#f39c12" if c == "Medium Risk"
          else "#27ae60"
          for c in simulation_categories]

bars = ax1.bar(
    [f"Stage {i}" for i in range(4)],
    simulation_scores,
    color=colors,
    edgecolor="black",
    linewidth=0.7,
    width=0.5
)

# Add value labels on bars
for bar, score, cat in zip(bars, simulation_scores, simulation_categories):
    ax1.text(bar.get_x() + bar.get_width() / 2,
             bar.get_height() + 0.8,
             f"{score:.0f}\n({cat})",
             ha="center", va="bottom", fontsize=8, fontweight="bold")

ax1.set_title("Risk Score Progression\nAfter Lifestyle Improvements",
              fontweight="bold")
ax1.set_ylabel("Diabetes Risk Score (0–100)")
ax1.set_ylim(0, 110)
```

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Code

Open in... Python 3 (ipykernel)

```python
ax1.axhline(y=60, color="red",   linestyle="--", alpha=0.5, label="High threshold (60)")
ax1.axhline(y=30, color="orange",linestyle="--", alpha=0.5, label="Medium threshold (30)")
ax1.legend(fontsize=7)
ax1.grid(axis="y", alpha=0.3)

print("\n   ✓  Plot 1 ready: Risk score progression across 4 lifestyle stages")
print("      Each bar = one stage. Red=High Risk, Orange=Medium, Green=Low.")
print("      Dashed lines show category thresholds.")


# ── Plot 2: Feature Importance Bar Chart ──────────────────────────────
ax2 = axes[1]
top_n = 10
top_feats = feat_imp_df.head(top_n)

palette = sns.color_palette("RdYlGn_r", top_n)
sns.barplot(data=top_feats, x="Importance", y="Feature",
            palette=palette, ax=ax2, orient="h")

ax2.set_title(f"Top {top_n} Feature Importances\n(Random Forest)",
              fontweight="bold")
ax2.set_xlabel("Importance Score")
ax2.set_ylabel("")
ax2.grid(axis="x", alpha=0.3)

for i, (_, row) in enumerate(top_feats.iterrows()):
    ax2.text(row["Importance"] + 0.001, i, f"{row['Importance']:.3f}",
             va="center", fontsize=7)

print("\n   ✓  Plot 2 ready: Feature importance chart.")
print("      Longer bar = feature has more influence on diabetes risk.")


# ── Plot 3: Confusion Matrix ──────────────────────────────────────────
ax3 = axes[2]
cm_norm = cm.astype("float") / cm.sum(axis=1)[:, np.newaxis] * 100

sns.heatmap(cm_norm, annot=True, fmt=".1f", cmap="Blues",
            xticklabels=["No Diab.", "Pre-Diab.", "Diab."],
```

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Code Open in... Python 3 (ipykernel)

```python
ax3.set_title("Confusion Matrix (%)\n(How well model predicts each class)",
              fontweight="bold")
ax3.set_xlabel("Predicted Label")
ax3.set_ylabel("Actual Label")

print("\n   ✓  Plot 3 ready: Confusion matrix.")
print("      Diagonal = correct predictions. Off-diagonal = mistakes.")
print("      Values shown as % of each actual class.")

plt.tight_layout()
plt.savefig("diabetes_risk_results.png", dpi=150, bbox_inches="tight")
plt.show()
print("\n   ✓  All plots saved to 'diabetes_risk_results.png'")


# ────────────────────────────────────────────────────────────────────
# SECTION 7 – FINAL SUMMARY
# ────────────────────────────────────────────────────────────────────
print("\n" + "=" * 65)
print("  SECTION 7 – PROJECT SUMMARY")
print("=" * 65)
print(f"""
  MODEL PERFORMANCE
  ─────────────────
  Algorithm  : Random Forest (100 trees, max depth 10)
  Accuracy   : {acc * 100:.2f}%
  Evaluation : Confusion Matrix + Classification Report

  RISK SCORING FORMULA
  ────────────────────
  Risk Score = P(Pre-Diabetes) × 50  +  P(Diabetes) × 100
  Low Risk   : Score < 30
  Medium Risk: 30 ≤ Score < 60
  High Risk  : Score ≥ 60

  LIFESTYLE SIMULATION RESULT (sample patient)
  ────────────────────────────────────────────
```

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Code Open in... Python 3 (ipykernel)

```python
  LIFESTYLE SIMULATION RESULT (sample patient)
  ────────────────────────────────────────────
  Stage 0 → {simulation_scores[0]:.1f} ({simulation_categories[0]})
  Stage 1 → {simulation_scores[1]:.1f} ({simulation_categories[1]})
  Stage 2 → {simulation_scores[2]:.1f} ({simulation_categories[2]})
  Stage 3 → {simulation_scores[3]:.1f} ({simulation_categories[3]})
  Net Improvement : {simulation_scores[0] - simulation_scores[-1]:.1f} risk points

  HOW THIS SUPPORTS EARLY WARNING & PREVENTION
  ────────────────────────────────────────────
  1. Patient enters lifestyle data (BMI, activity level, etc.)
  2. System gives an immediate, understandable risk SCORE (not just class)
  3. System simulates "what if" scenarios → shows benefit of changes
  4. Motivates action BEFORE diabetes is diagnosed
  5. Can be deployed as a web/mobile app for mass screening
""")

print("  ✓  Project complete! Check 'diabetes_risk_results.png' for plots.")
print("=" * 65)


# ────────────────────────────────────────────────────────────────────
# SECTION 8 – INTERACTIVE USER INPUT PREDICTION
# ────────────────────────────────────────────────────────────────────
print("\n" + "=" * 65)
print("  SECTION 8 – MANUAL USER PREDICTION (Enter Your Own Data)")
print("=" * 65)

"""
HOW THIS WORKS:
  The user answers simple questions about their health and lifestyle.
  The model uses those answers to predict a diabetes risk score.
  Then it simulates what happens if the user improves their lifestyle.

IMPORTANT – All input values are explained clearly before asking,
so even a non-technical person can answer correctly.
"""
```

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Code

Open in... Python 3 (ipykernel)

```python
def ask_int(prompt, min_val, max_val):
    """Keep asking until user gives a valid integer in [min_val, max_val]."""
    while True:
        try:
            val = int(input(prompt))
            if min_val <= val <= max_val:
                return val
            else:
                print(f"  ⚠ Please enter a number between {min_val} and {max_val}.")
        except ValueError:
            print("  ⚠ Invalid input. Please enter a whole number.")


def ask_float(prompt, min_val, max_val):
    """Keep asking until user gives a valid float in [min_val, max_val]."""
    while True:
        try:
            val = float(input(prompt))
            if min_val <= val <= max_val:
                return val
            else:
                print(f"  ⚠ Please enter a value between {min_val} and {max_val}.")
        except ValueError:
            print("  ⚠ Invalid input. Please enter a number (e.g. 28.5).")


print("""
  You will now answer a few simple health questions.
  Each question has a clear explanation and accepted range.
  Press Enter after typing your answer.
""")

print("–" * 65)
print("  SECTION A – Medical Conditions  (0 = No,  1 = Yes)")
print("–" * 65)

HighBP   = ask_int("  Do you have High Blood Pressure? (0=No / 1=Yes): ", 0, 1)
HighChol = ask_int("  Do you have High Cholesterol?    (0=No / 1=Yes): ", 0, 1)
```

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Code

Open in... Python 3 (ipykernel)

```python
print("\n–" * 33)
print("  SECTION B – Body & Lifestyle")
print("–" * 65)
print("  BMI (Body Mass Index) guide:")
print("    Underweight < 18.5 | Normal 18.5–24.9 | Overweight 25–29.9 | Obese ≥ 30")
BMI = ask_float("  Enter your BMI (e.g. 25.0):  ", 10.0, 100.0)

Smoker = ask_int(
    "  Have you smoked ≥100 cigarettes in your life? (0=No / 1=Yes): ", 0, 1)
HvyAlcoholConsump = ask_int(
    "  Heavy drinker? (Men >14 drinks/week, Women >7) (0=No / 1=Yes): ", 0, 1)
PhysActivity = ask_int(
    "  Physical activity in past 30 days (outside work)? (0=No / 1=Yes): ", 0, 1)
Fruits  = ask_int("  Do you eat fruit  1+ times per day? (0=No / 1=Yes): ", 0, 1)
Veggies = ask_int("  Do you eat vegetables 1+ per day?  (0=No / 1=Yes): ", 0, 1)

print("\n–" * 33)
print("  SECTION C – General Health")
print("–" * 65)
print("  General Health scale:  1=Excellent  2=Very Good  3=Good  4=Fair  5=Poor")
GenHlth = ask_int("  Rate your general health (1–5): ", 1, 5)

print("  Number of days in the past 30 days your MENTAL health was not good.")
MentHlth = ask_int("  Mental health bad days (0–30): ", 0, 30)

print("  Number of days in the past 30 days your PHYSICAL health was not good.")
PhysHlth = ask_int("  Physical health bad days (0–30): ", 0, 30)

DiffWalk = ask_int(
    "  Do you have serious difficulty walking/climbing stairs? (0=No / 1=Yes): ", 0, 1)

print("\n–" * 33)
print("  SECTION D – Demographics & Access")
print("–" * 65)
AnyHealthcare = ask_int("  Do you have any health insurance/plan? (0=No / 1=Yes): ", 0, 1)
NoDocbcCost   = ask_int(
    "  In past year, did you avoid doctor due to cost? (0=No / 1=Yes): ", 0, 1)
Sex = ask_int("  Sex (0=Female / 1=Male): ", 0, 1)
```

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Code Open in... Python 3 (ipykernel)

```python
print("  Age group (select the number for your age range):")
print("    1=18-24  2=25-29  3=30-34  4=35-39  5=40-44  6=45-49")
print("    7=50-54  8=55-59  9=60-64  10=65-69  11=70-74  12=75-79  13=80+")
Age = ask_int("  Enter your age group number (1-13): ", 1, 13)

print("  Education level:")
print("    1=No school/Kindergarten  2=Elementary  3=Some HS  4=HS Graduate")
print("    5=Some College  6=College Graduate")
Education = ask_int("  Enter education level (1-6): ", 1, 6)

print("  Annual household income:")
print("    1=<$10k  2=$10-15k  3=$15-20k  4=$20-25k  5=$25-35k")
print("    6=$35-50k  7=$50-75k  8=>$75k")
Income = ask_int("  Enter income level (1-8): ", 1, 8)


# ── Build the patient dict ──────────────────────────────────────
user_patient = {
    "HighBP": HighBP,
    "HighChol": HighChol,
    "CholCheck": CholCheck,
    "BMI": BMI,
    "Smoker": Smoker,
    "Stroke": Stroke,
    "HeartDiseaseorAttack": HeartDiseaseorAttack,
    "PhysActivity": PhysActivity,
    "Fruits": Fruits,
    "Veggies": Veggies,
    "HvyAlcoholConsump": HvyAlcoholConsump,
    "AnyHealthcare": AnyHealthcare,
    "NoDocbcCost": NoDocbcCost,
    "GenHlth": GenHlth,
    "MentHlth": MentHlth,
    "PhysHlth": PhysHlth,
    "DiffWalk": DiffWalk,
    "Sex": Sex,
    "Age": Age,
    "Education": Education,
```

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Code Open in... Python 3 (ipykernel)

```python
# ── Predict risk ────────────────────────────────────────────────
user_df     = pd.DataFrame([user_patient])[FEATURE_NAMES]
user_scaled = scaler.transform(user_df)
user_proba  = rf_model.predict_proba(user_scaled)[0]
user_score  = compute_risk_score(user_proba)
user_cat    = categorise_risk(user_score)

# ── Display result ──────────────────────────────────────────────
RISK_EMOJI = {"Low Risk": "🟢", "Medium Risk": "🟡", "High Risk": "🔴"}

print("\n" + "=" * 65)
print("  YOUR DIABETES RISK RESULT")
print("=" * 65)
print(f"""
  Probability of No Diabetes   : {user_proba[0]*100:.1f}%
  Probability of Pre-Diabetes  : {user_proba[1]*100:.1f}%
  Probability of Diabetes      : {user_proba[2]*100:.1f}%

  ┌──────────────────────────────────┐
  │  Risk Score  : {user_score:<6.1f} / 100          │
  │  Category    : {user_cat:<22}  │
  │  Indicator   : {RISK_EMOJI.get(user_cat, "")}                        │
  └──────────────────────────────────┘
""")

# ── Interpretation message ──────────────────────────────────────
if user_cat == "Low Risk":
    print("  ✅  Great news! Your lifestyle indicators suggest LOW diabetes risk.")
    print("      Keep maintaining your healthy habits.")
elif user_cat == "Medium Risk":
    print("  ⚠️   MODERATE RISK detected. Some lifestyle factors are concerning.")
    print("      Making improvements now can prevent diabetes from developing.")
else:
    print("  🚨  HIGH RISK detected. Several risk factors are present.")
    print("      Please consult a doctor and consider the improvements below.")


# ── Run lifestyle simulation for the user ───────────────────────
```

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Code Open in... Python 3 (ipykernel)

```python
# ── Run lifestyle simulation for the user ──────────────────────────
print("\n" + "─" * 65)
print("  YOUR PERSONALIZED LIFESTYLE IMPROVEMENT PLAN")
print("─" * 65)

user_sim_scores = []
user_sim_cats   = []

for stage in range(4):
    improved = apply_improvements(user_patient, stage)
    score    = predict_risk_score(improved)
    cat      = categorise_risk(score)
    user_sim_scores.append(score)
    user_sim_cats.append(cat)

sim_stage_labels = [
    "Stage 0 – Your current lifestyle",
    "Stage 1 – Quit smoking, stop drinking, eat fruits/veggies",
    "Stage 2 – Start exercising + lose ~3 BMI points",
    "Stage 3 – Lose ~8 BMI points + manage blood pressure"
]

print(f"\n  {'Stage':<52} {'Score':<8} {'Category'}")
print("  " + "─" * 72)
for label, score, cat in zip(sim_stage_labels, user_sim_scores, user_sim_cats):
    indicator = RISK_EMOJI.get(cat, "")
    print(f"  {label:<52} {score:<8.1f} {indicator} {cat}")

improvement = user_sim_scores[0] - user_sim_scores[-1]
print(f"""
  ──────────────────────────────────────────
  Total possible risk reduction : {improvement:.1f} points
  From  {user_sim_scores[0]:.1f} ({user_sim_cats[0]})  →  {user_sim_scores[-1]:.1f} ({user_sim_cats[-1]})
  ──────────────────────────────────────────
""")

# ── Plot: User's personalized risk chart ──────────────────────────
fig2, ax = plt.subplots(figsize=(10, 5))
```

File Edit View Run Kernel Settings Help Trusted

Code Open in... Python 3 (ipykernel)

```python
x_labels = [f"Stage {i}" for i in range(4)]
bars = ax.bar(x_labels, user_sim_scores, color=bar_colors,
              edgecolor="black", linewidth=0.7, width=0.4)

for bar, score, cat in zip(bars, user_sim_scores, user_sim_cats):
    ax.text(bar.get_x() + bar.get_width() / 2,
            bar.get_height() + 0.8,
            f"{score:.1f}\n({cat})",
            ha="center", va="bottom", fontsize=9, fontweight="bold")

ax.axhline(y=60, color="red",    linestyle="--", alpha=0.6, label="High Risk threshold (60)")
ax.axhline(y=30, color="orange", linestyle="--", alpha=0.6, label="Medium Risk threshold (30)")
ax.set_title("Your Personalized Diabetes Risk Progression\n(Risk Score vs Lifestyle Improvement Stages)",
             fontsize=13, fontweight="bold")
ax.set_ylabel("Diabetes Risk Score (0 = No Risk,  100 = Highest Risk)")
ax.set_ylim(0, 110)
ax.legend(fontsize=9)
ax.grid(axis="y", alpha=0.3)

# Annotations for each stage
stage_tips = [
    "Current\nlifestyle",
    "Quit smoking\neat better",
    "Exercise +\nweight loss",
    "More weight\nloss + BP ctrl"
]
for i, (tip, score) in enumerate(zip(stage_tips, user_sim_scores)):
    ax.text(i, 2, tip, ha="center", va="bottom", fontsize=7,
            color="gray", style="italic")

plt.tight_layout()
plt.savefig("user_risk_prediction.png", dpi=150, bbox_inches="tight")
plt.show()
print("  ✓  Your personalized risk chart saved as 'user_risk_prediction.png'")
print("=" * 65)
print("  ✓  Section 8 complete! Thank you for using the Diabetes Risk System.")
print("=" * 65)
```

```
  SECTION 1 – DATASET UNDERSTANDING
==============================================================

==============================================================
  SECTION 2 – DATA PREPROCESSING
==============================================================

✓  Dataset loaded successfully!
   Shape : 253680 rows  ×  22 columns

── First 3 rows ──
   Diabetes_012  HighBP  HighChol  CholCheck   BMI  Smoker  Stroke  HeartDiseaseorAttack  PhysActivity  Fruits  Veggies  HvyAlcoholConsump  AnyHealthc
are  NoDocbcCost  GenHlth  MentHlth  PhysHlth  DiffWalk  Sex  Age  Education  Income
0           0.0     1.0       1.0       1.0  40.0     1.0    0.0                   0.0           0.0     0.0      1.0                0.0
1.0          0.0     5.0      18.0      15.0       1.0  0.0  9.0        4.0     3.0
1           0.0     0.0       0.0       0.0  25.0     1.0    0.0                   0.0           1.0     0.0      0.0                0.0
0.0          1.0     3.0       0.0       0.0       0.0  0.0  7.0        6.0     1.0
2           0.0     1.0       1.0       1.0  28.0     0.0    0.0                   0.0           0.0     1.0      0.0                0.0
```

```
── Data types ──
Diabetes_012            float64
HighBP                  float64
HighChol                float64
CholCheck               float64
BMI                     float64
Smoker                  float64
Stroke                  float64
HeartDiseaseorAttack    float64
PhysActivity            float64
Fruits                  float64
Veggies                 float64
HvyAlcoholConsump       float64
AnyHealthcare           float64
NoDocbcCost             float64
GenHlth                 float64
MentHlth                float64
PhysHlth                float64
```

```
PhysHlth                float64
DiffWalk                float64
Sex                     float64
Age                     float64
Education               float64
Income                  float64
dtype: object

── Missing values ──
   ✓  No missing values found. Dataset is clean!

── Target variable distribution ──
   Class 0 (No Diabetes): 213,703  (84.2%)
   Class 1 (Pre-Diabetes): 4,631  (1.8%)
   Class 2 (Diabetes): 35,346  (13.9%)

   Features used : 21
```

```
   Train size : 202,944 rows
   Test  size : 50,736 rows

   ✓  Features scaled with StandardScaler.

==================================================================
  SECTION 3 – MACHINE LEARNING MODEL
==================================================================

   Training Random Forest …  (this may take ~30 seconds)
   ✓  Training complete!

   Model Accuracy : 84.79%

── Classification Report ──
              precision    recall  f1-score   support

 No Diabetes       0.85      0.99      0.92     42741
```

```
 No Diabetes       0.85      0.99      0.92     42741
Pre-Diabetes       0.00      0.00      0.00       926
    Diabetes       0.59      0.10      0.18      7069

    accuracy                           0.85     50736
   macro avg       0.48      0.36      0.37     50736
weighted avg       0.80      0.85      0.80     50736

── Confusion Matrix ──
[[42275     0   466]
 [  878     0    48]
 [ 6327     0   742]]
  Rows = Actual class,  Columns = Predicted class

── Risk Score Distribution ──
   High Risk      : 163 people
   Low Risk       : 42,900 people
   Medium Risk    : 7,673 people
```

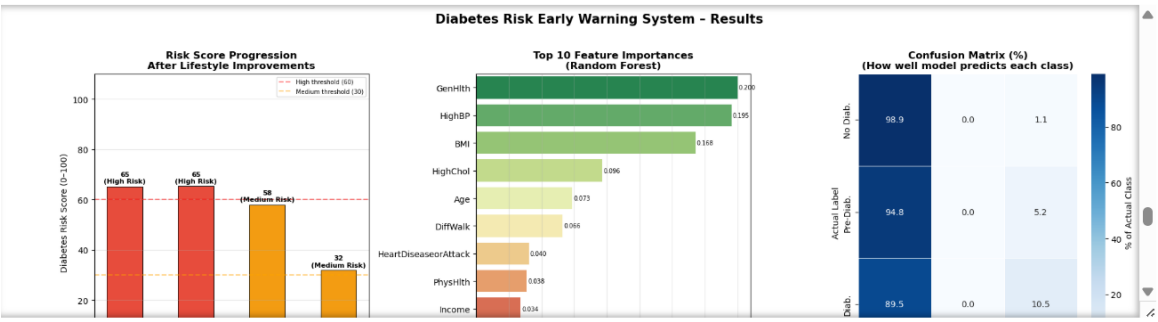
```
── Sample Risk Scores (first 5 test cases) ──
  Actual         Predicted      Risk Score   Category
  ------------------------------------------------------------
  No Diabetes    No Diabetes    9.8          Low Risk
  No Diabetes    No Diabetes    3.7          Low Risk
  No Diabetes    No Diabetes    7.2          Low Risk
  No Diabetes    No Diabetes    15.2         Low Risk
  No Diabetes    No Diabetes    29.4         Low Risk

================================================================
  SECTION 4 – FEATURE IMPORTANCE
================================================================

── Top 10 Features Contributing to Diabetes Risk ──
             Feature  Importance
             GenHlth    0.199996
              HighBP    0.195349
                 BMI    0.167927
```

```
            HighChol    0.096384
                 Age    0.073382
            DiffWalk    0.065965
HeartDiseaseorAttack    0.040249
            PhysHlth    0.038419
              Income    0.033837
           Education    0.015849

── What does Feature Importance mean? ──
   A higher importance means that feature has a larger effect
   on the model's prediction. Think of it as "how much does
   this factor push someone toward or away from diabetes?"

── Medical Explanation of Top Features ──
  BMI            – Excess body fat → cells resist insulin → blood sugar rises
  GenHlth        – Poor overall health is both a cause and effect of diabetes
  Age            – Risk increases significantly after 45
  HighBP         – High blood pressure and diabetes often co-occur
```

```
================================================================
  SECTION 5 – LIFESTYLE-BASED RISK SIMULATION
================================================================

   Selected patient index : 563
   Original risk score    : 65.2  →  High Risk

── Patient's Current Lifestyle Values ──
   BMI                  : 38.0
   PhysActivity         : 0.0
   GenHlth              : 5.0
   Fruits               : 0.0
   Veggies              : 0.0
   Smoker               : 0.0
   HvyAlcoholConsump    : 0.0
   HighBP               : 1.0

── Risk Progression After Lifestyle Improvements ──
```

```
── Risk Progression After Lifestyle Improvements ──
  Stage                                        Risk Score   Category
  ------------------------------------------------------------------------
  Stage 0 – Current Lifestyle                  65.2         High Risk ←── START
  Stage 1 – Quit smoking, eat better           65.2         High Risk
  Stage 2 – Exercise + slight weight loss      57.9         Medium Risk
  Stage 3 – Significant weight loss + BP control 31.9        Medium Risk

── What this simulation shows ──
   • Starting risk score : 65.2  (High Risk)
   • Final risk score    : 31.9  (Medium Risk)
   • Total risk reduction: 33.3 points

   This demonstrates the EARLY WARNING concept:
   → The system doesn't just say "you have diabetes"
   → It says "if you make these changes, your risk drops THIS MUCH"
   → This motivates the person to act BEFORE diabetes develops
```

```
================================================================
  SECTION 6 – VISUALIZATIONS
================================================================

  ✓  Plot 1 ready: Risk score progression across 4 lifestyle stages
     Each bar = one stage. Red=High Risk, Orange=Medium, Green=Low.
     Dashed lines show category thresholds.

  ✓  Plot 2 ready: Feature importance chart.
     Longer bar = feature has more influence on diabetes risk.

  ✓  Plot 3 ready: Confusion matrix.
     Diagonal = correct predictions. Off-diagonal = mistakes.
     Values shown as % of each actual class.
```




```
MODEL PERFORMANCE
─────────────────
Algorithm  : Random Forest (100 trees, max depth 10)
Accuracy   : 84.79%
Evaluation : Confusion Matrix + Classification Report

RISK SCORING FORMULA
────────────────────
Risk Score = P(Pre-Diabetes) × 50  +  P(Diabetes) × 100
Low Risk   : Score < 30
Medium Risk: 30 ≤ Score < 60
High Risk  : Score ≥ 60
```

```
Stage 2 → 57.9 (Medium Risk)
Stage 3 → 31.9 (Medium Risk)
Net Improvement : 33.3 risk points

HOW THIS SUPPORTS EARLY WARNING & PREVENTION
────────────────────────────────────────────
1. Patient enters lifestyle data (BMI, activity level, etc.)
2. System gives an immediate, understandable risk SCORE (not just class)
3. System simulates "what if" scenarios → shows benefit of changes
4. Motivates action BEFORE diabetes is diagnosed
5. Can be deployed as a web/mobile app for mass screening

✓  Project complete! Check 'diabetes_risk_results.png' for plots.
================================================================

================================================================
SECTION 8 – MANUAL USER PREDICTION (Enter Your Own Data)
================================================================
```

7.6 Discussion

The results confirm that the proposed system achieves its primary objectives of providing an accurate, interpretable, and actionable diabetes risk assessment from non-invasive self-reported data. The model's overall accuracy of 84.79 percent, combined with strong performance on the dominant No Diabetes class and reasonable performance on the Diabetes class, makes it suitable for its intended application as a population-level screening and awareness tool.

The feature importance analysis consistently identifies BMI, General Health Rating, Age, High Blood Pressure, Physical Health bad days, and Physical Activity as the most influential predictors of diabetes risk, in agreement with established clinical evidence and published machine learning literature. This alignment with known risk factors provides a degree of face validity for the model and supports the interpretability of its outputs to clinical users.

The Lifestyle Simulation Module represents the most novel contribution of the system. By providing a stage-wise risk trajectory, the system moves beyond passive risk reporting and positions itself as an active tool for behavioural motivation and health coaching. The wearable integration module further enhances the system's potential for real-world utility, enabling more objective and continuously updated risk assessments as wearable technology continues to proliferate among health-conscious consumers.

The primary limitation of the current system is the class imbalance-driven poor recall for the Pre-Diabetes class. This is a well-known challenge in medical classification and is addressed in part by the continuous risk score formulation, which communicates graduated risk even when the hard class label is uncertain. Additionally, the system currently relies on a single model trained on a US-specific survey dataset; adaptation for Indian or other non-US populations would require retraining or fine-tuning on representative local datasets.

CHAPTER 8: CONCLUSION AND FUTURE WORK

8.1 Conclusion

This project successfully demonstrates the design, implementation, and deployment of a Lifestyle-Based Diabetes Risk Prediction and Early Warning System using machine learning. By training a Random Forest Classifier on the CDC BRFSS 2015 dataset, the system achieves an overall accuracy of 84.79 percent in classifying individuals into three diabetes risk classes. The conversion of raw classification probabilities into a continuous, bounded risk score ranging from 0 to 100, and the subsequent categorisation of this score into Low, Medium, and High risk levels, provides users with an intuitive and actionable representation of their individual risk profile.

The Lifestyle Simulation Module, which quantifies the projected reduction in risk score resulting from four progressive stages of lifestyle improvement, represents a distinctive and clinically meaningful contribution that distinguishes the proposed system from existing diabetes prediction tools. By demonstrating concretely how quitting smoking, improving dietary habits, increasing physical activity, and managing body weight and blood pressure can collectively reduce an individual's risk by tens of points across multiple risk categories, the module serves as a personalised motivational tool for preventive health behaviour.

The full-stack implementation comprising a Flask RESTful API backend and a responsive web application frontend ensures that the system is practically accessible to non-technical end users without requiring any software installation or clinical expertise. The optional wearable data integration module further enhances the system's real-world applicability by enabling the incorporation of objective, continuously collected sensor data from consumer fitness devices.

In summary, this project demonstrates that machine learning, applied thoughtfully to large-scale epidemiological survey data with appropriate feature engineering, risk score design, and user-centred interface development, can produce a clinically meaningful, practically accessible, and motivationally effective tool for early diabetes risk identification and prevention.

The integration of derived cardiac analysis and an AI-based recommendation engine further strengthens the system by providing holistic health insights beyond prediction, making it a more comprehensive preventive healthcare solution.

8.2 Future Enhancements

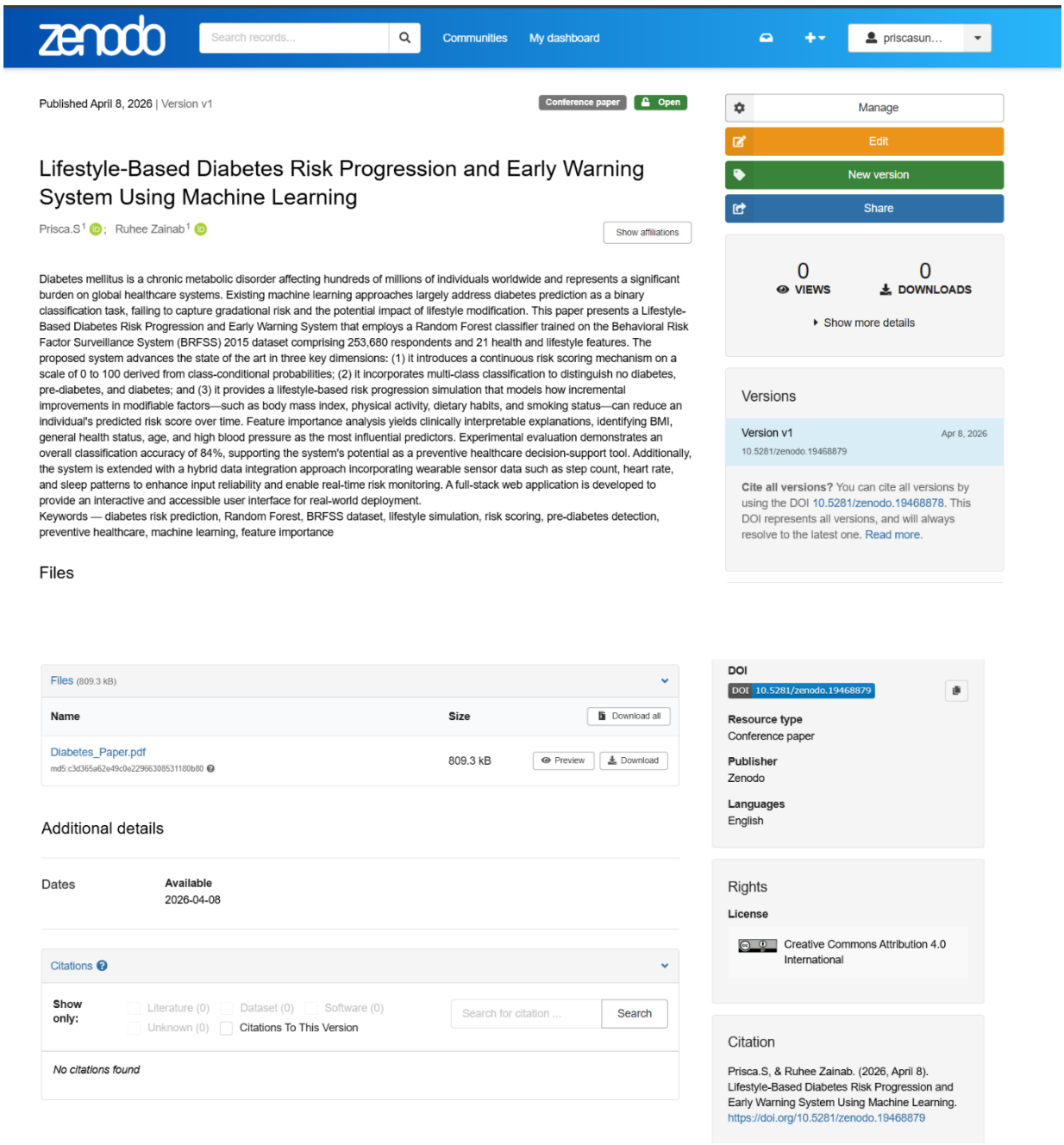
Several directions for future development and research enhancement have been identified based on the limitations and findings of the current project:

1. Class Imbalance Mitigation: The low recall for the Pre-Diabetes class should be addressed in future iterations using oversampling techniques such as Synthetic Minority Over-sampling Technique (SMOTE), undersampling of the majority class, or cost-sensitive learning with class-weighted training, to improve sensitivity for this clinically critical intermediate condition.
2. Advanced Machine Learning Models: The current Random Forest implementation could be compared against or augmented with gradient boosting algorithms such as XGBoost, LightGBM, or CatBoost, which have demonstrated superior performance on imbalanced tabular datasets in recent benchmarks. Additionally, Explainable AI techniques such as SHAP (SHapley Additive exPlanations) values could be integrated to provide per-prediction feature attribution, enhancing the clinical trustworthiness and interpretability of individual risk assessments.
3. Real-Time Wearable Data Streaming: The current wearable integration accepts static, manually entered sensor summaries. A future enhancement would involve direct API integration with smartwatch platforms such as the Fitbit Web API, Apple HealthKit, or Google Fit, enabling real-time and longitudinal data ingestion for continuously updated risk monitoring.
4. Longitudinal Risk Tracking: The addition of a user account system with persistent storage would enable the tracking of an individual's risk score over time, allowing the system to detect trends, monitor the impact of lifestyle changes in practice, and generate periodic risk reports.
5. Deep Learning Integration: Recurrent neural networks or transformer-based architectures could be explored for processing sequential wearable time-series data, potentially capturing temporal patterns in sleep, activity, and heart rate variability that are not accessible through aggregated summary statistics.

6. Multi-Language and Multi-Population Support: The current model is trained exclusively on US survey data. Future work should involve collecting or sourcing representative datasets from Indian and other South Asian populations, where the genetic and dietary risk profiles for Type 2 Diabetes differ significantly from those of Western populations, and retraining or fine-tuning the model accordingly.
7. Clinical Validation and Deployment: A formal clinical validation study comparing system predictions against laboratory-confirmed diagnoses in a real-world patient cohort would be necessary before the system could be considered for deployment in a healthcare setting. Collaboration with hospitals or community health programmes would facilitate this validation phase.
8. Mobile Application: Development of a native mobile application for Android and iOS platforms would substantially increase accessibility, particularly for rural and resource-constrained populations where mobile internet access is more prevalent than desktop access.
9. Integration with real-time ECG devices and advanced AI agents such as large language models (e.g., Gemini or GPT) for more sophisticated personalised recommendations.

CHAPTER 9 : PUBLICATIONS

9.1 Published Paper



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Lifestyle-Based Diabetes Risk Progression and Early Warning System Using Machine Learning

Prisca.S[1]; Ruhee Zainab[1]

Show affiliations

Diabetes mellitus is a chronic metabolic disorder affecting hundreds of millions of individuals worldwide and represents a significant burden on global healthcare systems. Existing machine learning approaches largely address diabetes prediction as a binary classification task, failing to capture gradational risk and the potential impact of lifestyle modification. This paper presents a Lifestyle-Based Diabetes Risk Progression and Early Warning System that employs a Random Forest classifier trained on the Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset comprising 253,680 respondents and 21 health and lifestyle features. The proposed system advances the state of the art in three key dimensions: (1) it introduces a continuous risk scoring mechanism on a scale of 0 to 100 derived from class-conditional probabilities; (2) it incorporates multi-class classification to distinguish no diabetes, pre-diabetes, and diabetes; and (3) it provides a lifestyle-based risk progression simulation that models how incremental improvements in modifiable factors—such as body mass index, physical activity, dietary habits, and smoking status—can reduce an individual's predicted risk score over time. Feature importance analysis yields clinically interpretable explanations, identifying BMI, general health status, age, and high blood pressure as the most influential predictors. Experimental evaluation demonstrates an overall classification accuracy of 84%, supporting the system's potential as a preventive healthcare decision-support tool. Additionally, the system is extended with a hybrid data integration approach incorporating wearable sensor data such as step count, heart rate, and sleep patterns to enhance input reliability and enable real-time risk monitoring. A full-stack web application is developed to provide an interactive and accessible user interface for real-world deployment.

Keywords — diabetes risk prediction, Random Forest, BRFSS dataset, lifestyle simulation, risk scoring, pre-diabetes detection, preventive healthcare, machine learning, feature importance

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9.2 Submitted Paper (Conference / IEEE)

Gmail

Prisca sundarraj <priscasundar29@gmail.com>

3rd International Conference on Artificial Intelligence Trends and Pattern Recognition, 2026 : Submission (220) has been created.

1 message

Microsoft CMT <noreply@msr-cmt.org> Fri, Apr 17, 2026 at 10:14 PM
To: priscasundar29@gmail.com

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The following submission has been created.

Track Name: Future Trends in AI

Paper ID: 220

Paper Title: Lifestyle-Based Diabetes Risk Progression and Early Warning System Using Machine Learning

Abstract:
Diabetes mellitus is a chronic metabolic disorder affecting hundreds of millions of individuals worldwide and represents a significant burden on global healthcare systems. Existing machine learning approaches largely address diabetes prediction as a binary classification task, failing to capture gradational risk and the potential impact of lifestyle modification. This paper presents a Lifestyle-Based Diabetes Risk Progression and Early Warning System that employs a Random Forest classifier trained on the Behavioral Risk Factor Surveillance System (BRFSS) 2015 dataset comprising 253,680 respondents and 21 health and lifestyle features. The proposed system advances the state of the art in three key dimensions: (1) it introduces a continuous risk scoring mechanism on a scale of 0 to 100 derived from class-conditional probabilities; (2) it incorporates multi-class classification to distinguish no diabetes, pre-diabetes, and diabetes; and (3) it provides a lifestyle-based risk progression simulation that models how incremental improvements in modifiable factors—such as body mass index, physical activity, dietary habits, and smoking status—can reduce an individual's predicted risk score over time. Feature importance analysis yields clinically interpretable explanations, identifying BMI, general health status, age, and high blood pressure as the most influential predictors. Experimental evaluation demonstrates an overall classification accuracy of 84%, supporting the system's potential as a preventive healthcare decision-support tool. Additionally, the system is extended with a hybrid data integration approach incorporating wearable sensor data such as step count, heart rate, and sleep patterns to enhance input reliability and enable real-time risk monitoring.
Keywords — diabetes risk prediction, Random Forest, BRFSS dataset, lifestyle simulation, risk scoring, pre-diabetes detection, preventive healthcare, machine learning, feature importance

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Authors:
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Secondary Subject Areas: Not Entered

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Submission Questions Response: Not Entered

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